Investigating the Water Use Efficiency of the Major Crops in Arid and Semi-arid regions through Geospatial Techniques

1. Introduction

Water is a vital source of life on earth and it is very important in the agricultural productivity. It forms the core part of the plant growth which influences the crop yields, soil health and the whole cycle of the crops. The Food and Agriculture Organization (FAO, 2020) estimated that 70 percent of freshwater is consumed in the world in agriculture, which is an indication of the significance of effective water management in the agricultural processes. Effective water use is even more necessary in Pakistan due to a scarcity of water resources that are inconsistently distributed. Irrigation, especially in the arid and semi-arid areas, is central as the country relies on it to eliminate the problem of food insecurity and agricultural sustainability (Hussain et al., 2022). Thus, the mechanisms of water utilization and improvement of its use become the key to ensuring the health of agriculture development and sustainability particularly in zones with water scarcity problem.

Agriculture plays a crucial role in the economic development of Pakistan, as it not only contributes significantly to the GDP but also provides employment to a substantial portion of the population. As of 2020, agriculture contributes around 18.9% to the national GDP and employs about 42.3% of the total workforce (Khan et al., 2021). This sector is pivotal for the country's food security, rural livelihoods, and export revenue. The country's agricultural landscape is diverse, with both subsistence farming and large-scale commercial farming. The province of Punjab, which is the country's agricultural heartland, accounts for a major portion of the nation's crop production. However, despite its importance, agriculture in Pakistan faces various challenges such as water scarcity, climate change, and inefficient irrigation practices. Addressing these issues is essential for enhancing agricultural productivity and achieving long-term sustainable development.

Punjab, often referred to as the "breadbasket" of Pakistan, is home to a wide variety of crops, which include staple food grains, fruits, and cash crops. The province's fertile soil and abundant irrigation network, primarily sourced from the Indus River system, support the cultivation of key crops such as wheat, rice, cotton, and sugarcane (Ali et al., 2019). Wheat and rice are the primary food crops, with Punjab contributing nearly 80% of Pakistan's rice production and over 50% of wheat production (Shah et al., 2021). Cotton and sugarcane, on the other hand, are important cash crops that contribute significantly to Pakistan's agricultural exports. Despite the large-scale production of these crops, challenges such as water availability, soil salinity, and inefficient irrigation methods continue to affect crop yields and the sustainability of farming practices in the region.

Evapotranspiration (ET) is a critical process in the hydrological cycle, involving the combined effects of evaporation and plant transpiration. Evaporation refers to the loss of water from soil and water bodies into the atmosphere, while transpiration involves the movement of water through plants. ET plays a vital role in understanding the water requirements of crops and is essential for efficient water management in agriculture (Mitra et al., 2020). ET is affected by various factors such as climate, crop type, soil properties, and water availability. In regions like Punjab, where irrigation is heavily relied upon, accurate estimation of ET is crucial for optimizing water use and ensuring that crops receive the necessary amount of water for growth.

Crop water requirement (ETa) refers to the amount of water needed by a crop to reach its full potential yield, considering all environmental factors (Jensen et al., 2018). ETa is determined by multiplying the reference evapotranspiration (ETo) with a crop coefficient (Kc) that reflects the specific needs of the crop. Accurate estimation of ETa is essential for efficient irrigation planning and water management. In Punjab, where agricultural water use is already under stress, understanding the crop water requirement helps in optimizing irrigation schedules and preventing both over-irrigation and under-irrigation, which can lead to water wastage or crop stress.

Reference evapotranspiration (ETo) is the amount of evapotranspiration that would occur from a hypothetical grass reference crop under specified conditions (Allen et al., 2019). ETo is a useful parameter for estimating the water needs of different crops and is influenced by weather parameters such as temperature, humidity, wind speed, and solar radiation. Meteorological stations across Pakistan, including those in Punjab, routinely measure ETo to assist in irrigation management and agricultural planning (Khan et al., 2021). This data, along with other climatic variables, helps in predicting the amount of water needed for various crops at different stages of growth, contributing to more efficient water resource management.

The crop coefficient (Kc) is a dimensionless value used to adjust the reference evapotranspiration (ETo) to account for the specific water requirements of different crops (Allen et al., 2019). Kc varies throughout the crop growth cycle, with different values for different growth stages such as the initial, mid-season, and late-season stages. By incorporating Kc into irrigation scheduling, farmers can apply water more efficiently, ensuring that crops receive the optimal amount of water at the right time. Understanding and calculating Kc values for major crops in Punjab, such as wheat, rice, and cotton, is essential for improving water use efficiency and maximizing crop yields.

Water use efficiency (WUE) refers to the amount of crop yield produced per unit of water applied. Improving WUE is crucial for optimizing water resources, particularly in regions facing water scarcity (Mitra et al., 2020). In Pakistan, where water availability is becoming increasingly limited, enhancing WUE through better irrigation practices, crop selection, and technological interventions is essential for achieving sustainable agricultural productivity. WUE can be improved by adopting efficient irrigation systems such as drip irrigation, which minimizes water wastage, and by using crop varieties that are more drought-resistant or tolerant to water stress (Hussain et al., 2022).

Supervised classification techniques, such as the Random Forest algorithm, are used in remote sensing to classify land cover and crop types based on satellite imagery. Random Forest is an ensemble learning method that uses multiple decision trees to classify data into distinct categories. In agriculture, this technique is particularly useful for mapping crop types, estimating crop area, and monitoring crop health. Remote sensing data, combined with machine learning algorithms like Random Forest, can provide valuable insights into crop patterns and their water requirements, which can be used for better management of agricultural resources (Zhang et al., 2021).

Remote sensing involves the use of satellite and aerial imagery to monitor and analyze the Earth's surface. In agriculture, remote sensing provides valuable data on crop health, soil moisture, and land use, which can be used to improve water management and optimize irrigation practices. Remote sensing techniques, including optical and thermal imagery, can help in assessing evapotranspiration rates, identifying water stress, and monitoring crop growth (Mitra et al., 2020). In Punjab, remote sensing data has been increasingly used for crop monitoring and to assess water use efficiency, enabling farmers and policymakers to make data-driven decisions for sustainable agricultural development.

Geographic Information Systems (GIS) and Google Earth Engine (GEE) are powerful tools for spatial data analysis and mapping in agriculture. GIS enables the integration of various data sources, such as climate, soil, and crop data, into a single platform for spatial analysis (Liu et al., 2022). By combining GIS with remote sensing, agricultural researchers can model crop growth, monitor water use, and predict yield outcomes under different environmental scenarios. Google Earth Engine (GEE) is a cloud-based platform that provides access to large datasets of satellite imagery and geospatial data, which can be analyzed for monitoring environmental changes, assessing crop conditions, and improving water management (Gorelick et al., 2017). In Pakistan, GEE has been increasingly used for large-scale agricultural monitoring, particularly for tracking water availability and crop health in regions like Punjab.

2. Problem Statement

In arid and semi-arid regions, the efficient use of water is crucial for ensuring sustainable agriculture, yet the lack of precise evapotranspiration (ET) estimates and comprehensive studies on water use efficiency (WUE) for major crops presents significant challenges. This is especially true for areas like Faisalabad and Rahim Yar Khan, where water resources are already under pressure due to population growth, climate change, and shifting agricultural practices. Current water management practices fail to address the spatiotemporal variability in crop water use, leading to inefficiencies, reduced productivity, and environmental stress. Additionally, the absence of localized data on ET and WUE hampers the development of tailored, effective water management strategies. This study aims to fill this gap by using geospatial techniques to compute evapotranspiration and actual water use for major crops in these regions, analyzing the variability of WUE. The findings will provide vital insights for optimizing water management and improving crop productivity under changing climatic conditions.

3. Study Area

The study was conducted in two major agricultural districts of Punjab province, Pakistan: Faisalabad and Rahim Yar Khan. Faisalabad is located at 31.43°N latitude and 71.10°E longitude, while Rahim Yar Khan is situated at 28.38°N latitude and 70.38°E longitude, as depicted in Figure 3.1. These regions were selected due to their arid to semi-arid climatic conditions and their significance in crop cultivation, making them ideal for calculating evapotranspiration and analyzing the spatiotemporal variability of water use efficiency (WUE). The climate of these regions is characterized by hot summers, mild winters, and relatively low rainfall. In Faisalabad, the average minimum and maximum temperatures range from 15°C to 40°C, with extreme temperatures often exceeding 45°C in summer. The average annual rainfall is approximately 300-

350 mm, primarily concentrated during the monsoon season (June to September). In Rahim Yar Khan, temperatures can reach up to 50°C in summer, with minimum temperatures around 20°C in winter. The average annual rainfall here is slightly lower, around 200-250 mm, with most precipitation occurring during the monsoon months. Both regions have diverse soil types, with a predominance of loam and clay loam soils, which are typically suitable for growing a wide range of crops. The soil texture in these areas influences water retention and drainage characteristics, which are critical factors in determining water use efficiency. The soil in Rahim Yar Khan tends to be slightly more saline due to high evaporation rates and groundwater abstraction, while Faisalabad's soil generally has better water-holding capacity. Together, these regions provide a comprehensive setting to study water use dynamics and crop productivity under varying climatic and soil conditions.

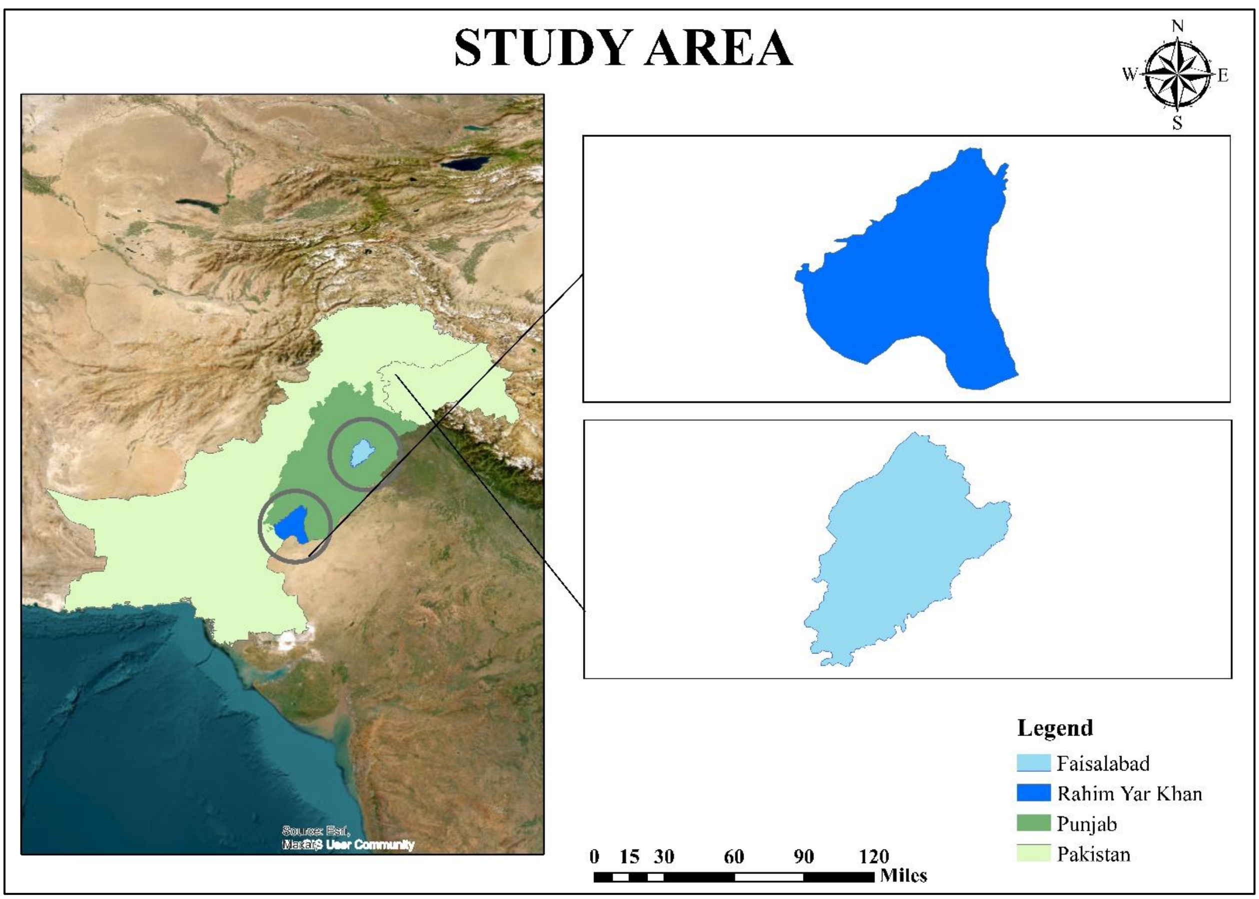
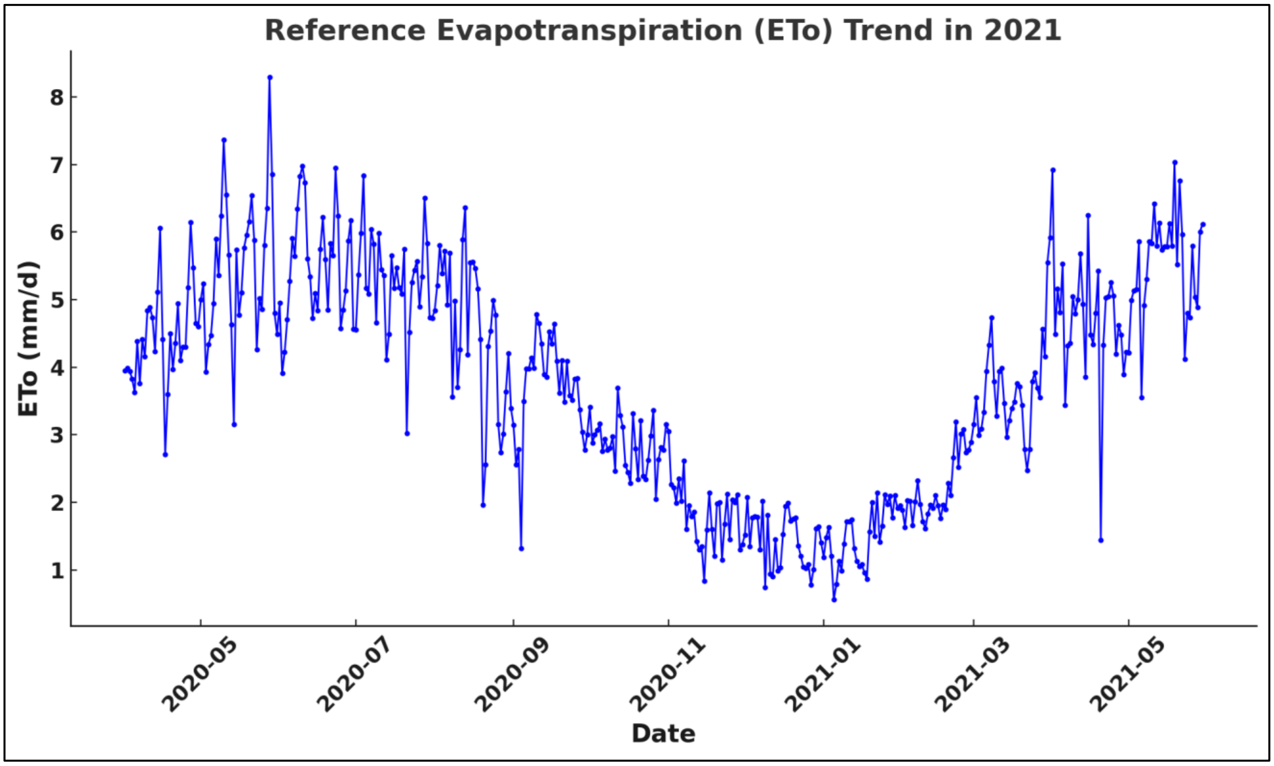


Figure 3.1. Geographical Map of Study Area

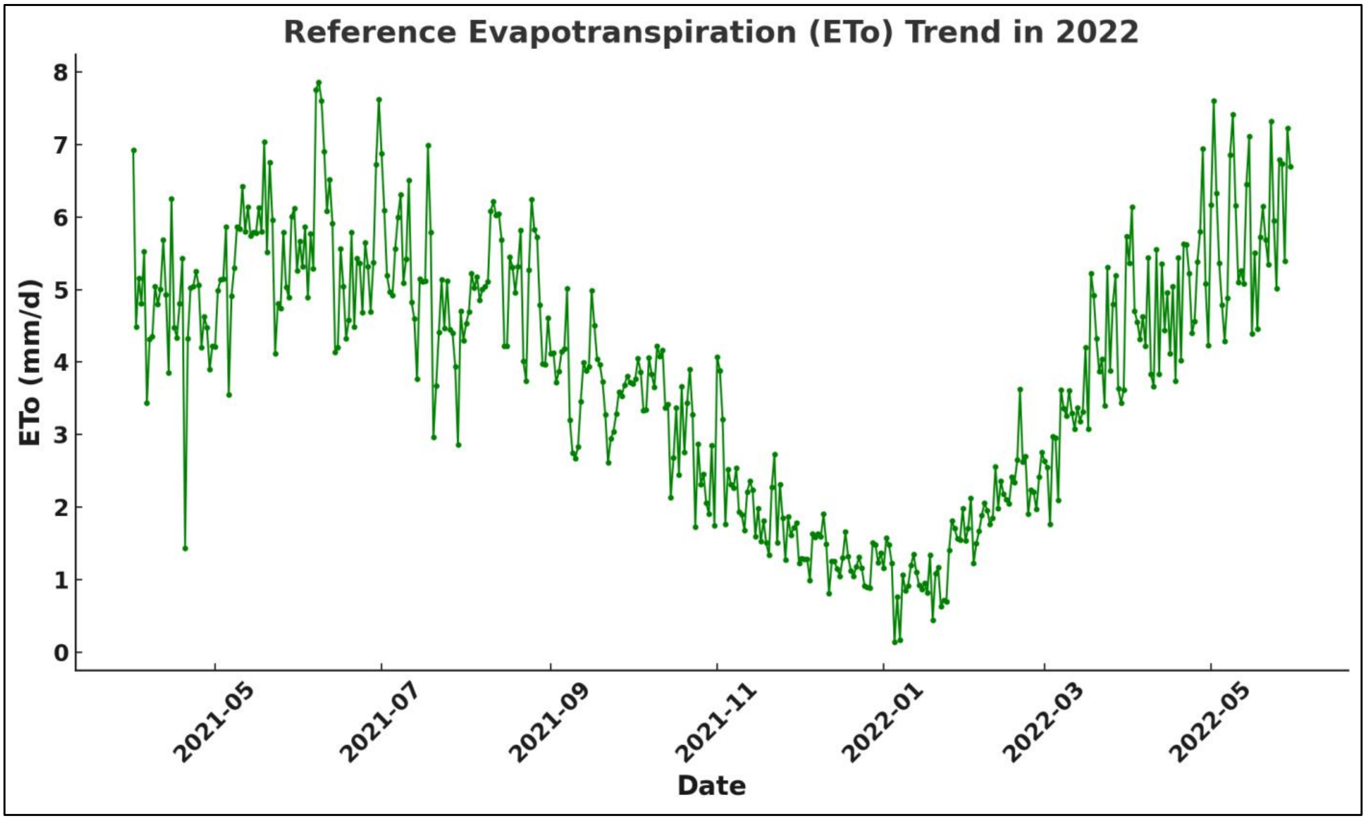
4. Reference Evapotranspiration

4.1. Reference Evapotranspiration of Faisalabad



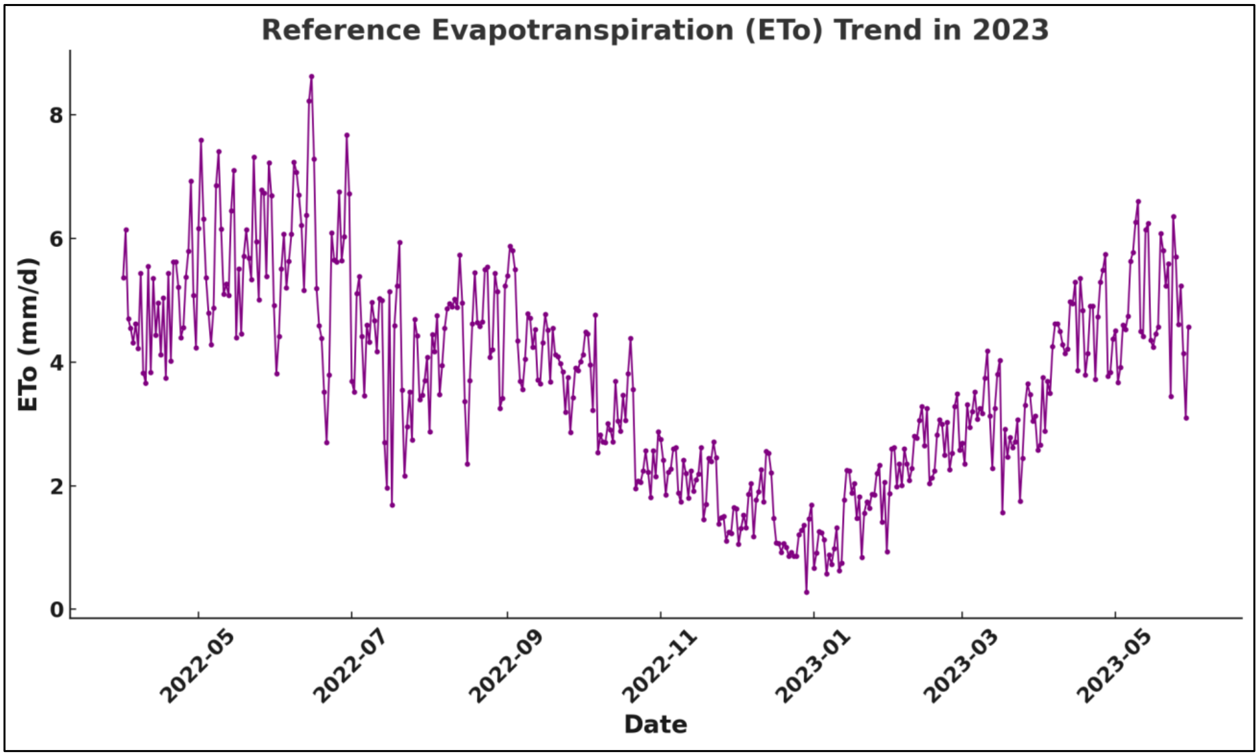
Graph 4.1 Reference Evapotranspiration of Faisalabad in 2021

The Reference Evapotranspiration (ET_o) Trend in 2021 reveals significant seasonal variations in evapotranspiration values, with both maximum and minimum values occurring at distinct points throughout the year. The highest ET_o values are observed during the peak summer months, particularly from May 19, 2021 to May 21, 2021, with values reaching up to 8.30 mm/day. This peak reflects the heightened evapotranspiration rates due to high temperatures and increased solar radiation, which drive higher evaporation rates. Conversely, the lowest ET_o values occur in the cooler months, particularly from November 2020 to February 2021, with the lowest recorded ET_o of 0.56 mm/day on January 5, 2021. These reduced values are indicative of lower evaporation rates due to cooler temperatures and less intense sunlight. The seasonal variation in ET_o underlines the critical role of temperature, sunlight, and seasonal shifts in water use efficiency, which can aid in optimizing water management practices in agriculture throughout the year.



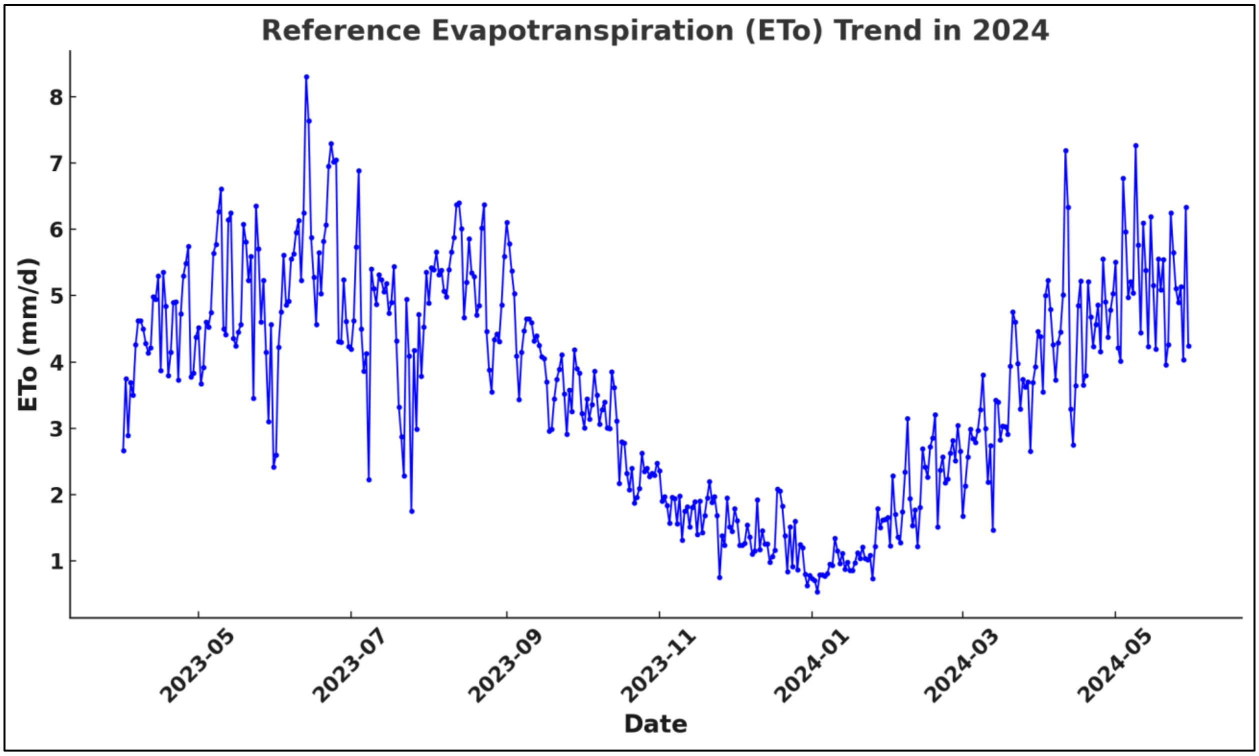
Graph 4.2 Reference Evapotranspiration of Faisalabad in 2022

The Reference Evapotranspiration (ET_o) Trend in 2021-2022 depicted in the graph shows significant fluctuations in daily evapotranspiration values (mm/day) over the course of the year. The maximum ET_o value for this period is 7.86 mm/day, recorded on June 8, 2021, marking the peak of evapotranspiration during the summer months due to high temperatures and solar radiation. Conversely, the minimum ET_o value is 0.14 mm/day, observed on January 5, 2022, during the colder winter months, where evaporation rates are lower due to reduced temperatures and sunlight. These extreme values highlight the seasonal nature of evapotranspiration, with higher rates in the warmer months and lower rates during the winter. This seasonal pattern is crucial for optimizing irrigation and water resource management, as it directly impacts water demand for crops across different seasons.



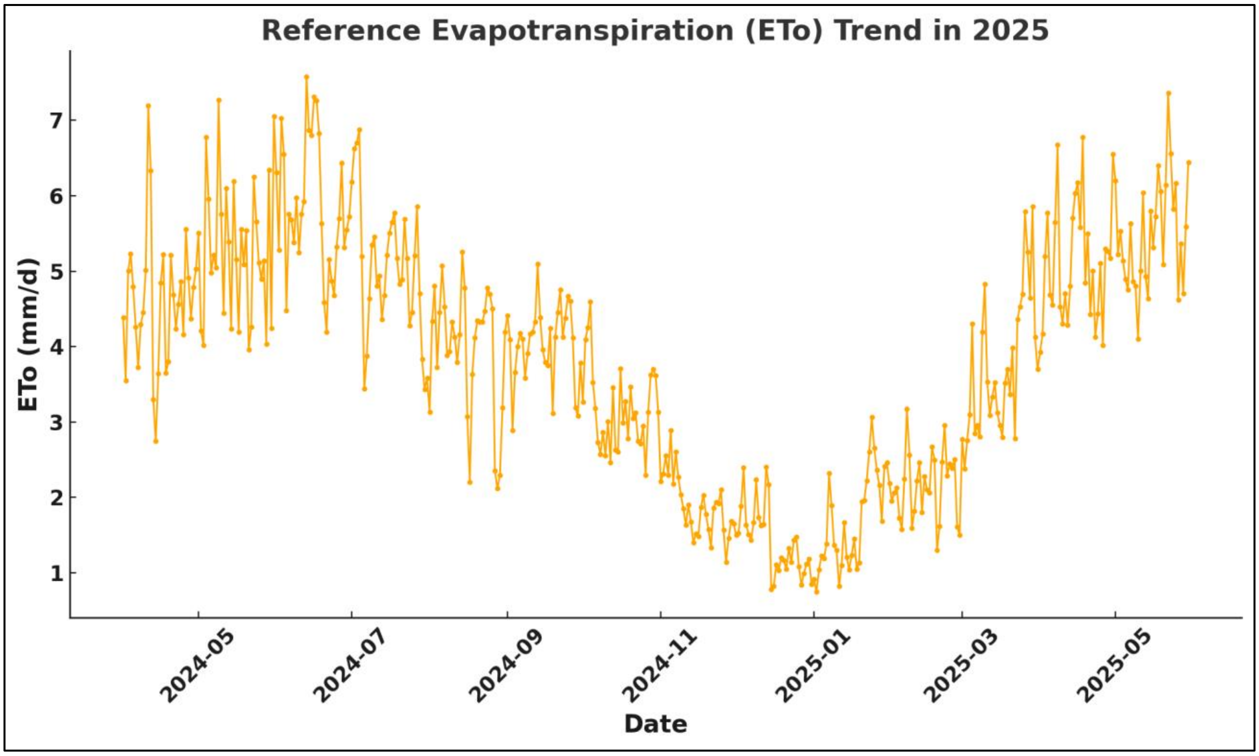
Graph 4.3 Reference Evapotranspiration of Faisalabad in 2023

The Reference Evapotranspiration (ET_o) Trend in 2022-2023 reveals significant seasonal fluctuations in evapotranspiration. The maximum ET_o value recorded during this period is 8.63 mm/day, observed on June 15, 2022, which corresponds to the peak of the summer season when higher temperatures and intense solar radiation lead to elevated evaporation rates. On the other hand, the minimum ET_o value is 0.28 mm/day, recorded on December 29, 2022, reflecting the reduced evaporation during the cooler winter months with lower temperatures and less sunlight. These variations highlight the critical role of temperature and sunlight in determining water consumption by crops, providing valuable insights for optimizing irrigation and water management strategies throughout the year.



Graph 4.4 Reference Evapotranspiration of Faisalabad in 2024

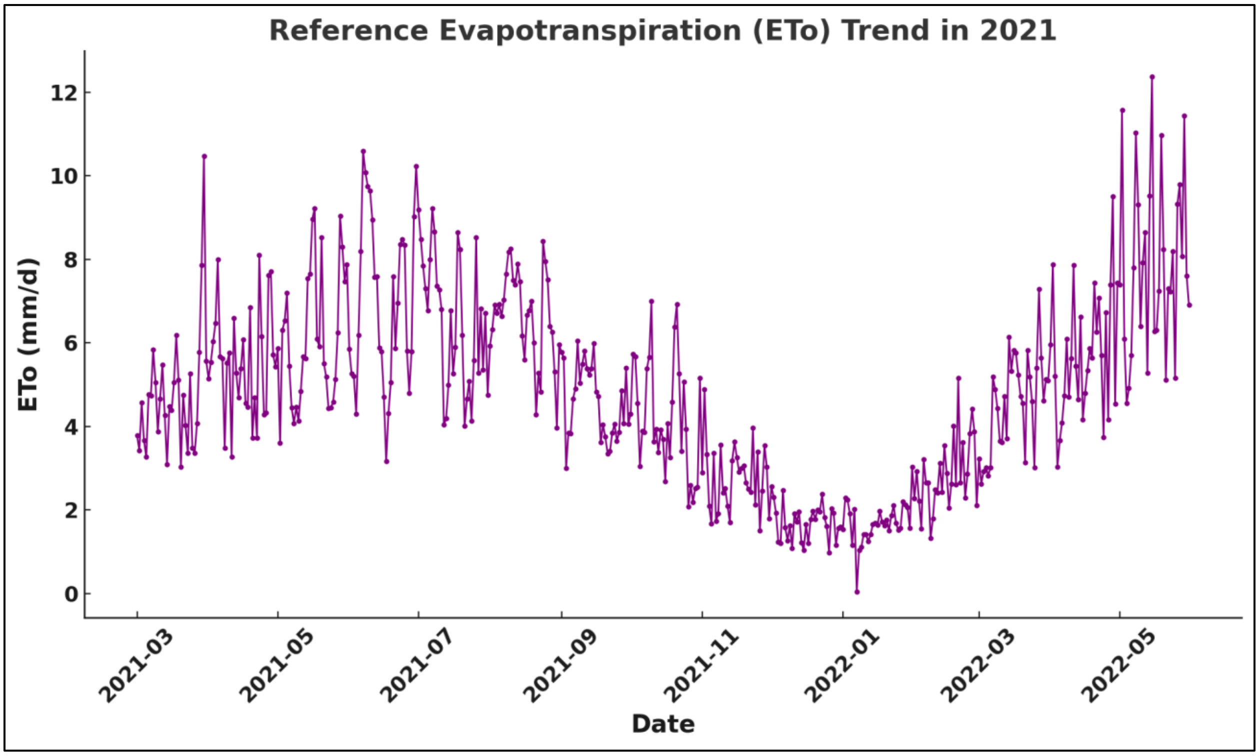
The Reference Evapotranspiration (ET_o) Trend in 2024 graph demonstrates clear seasonal fluctuations in evapotranspiration values throughout the year. The highest ET_o values occur during the warmer months, particularly from May to August 2024, where the evaporation rates rise significantly due to increased temperatures and solar radiation, peaking around 8 mm/day. In contrast, the lowest values are observed during the cooler months, particularly from December 2023 to February 2024, where evapotranspiration decreases to below 2 mm/day due to reduced temperatures and less sunlight. The interpolation has smoothed out the daily fluctuations, presenting a clear seasonal pattern of higher water demand during the summer and reduced water consumption in the winter. This trend emphasizes the critical role of temperature and sunlight in determining the water consumption of crops and highlights the importance of adapting irrigation schedules and water resource management strategies to these seasonal variations to ensure optimal crop growth and sustainable water use.



Graph 4.5 Reference Evapotranspiration of Faisalabad in 2025

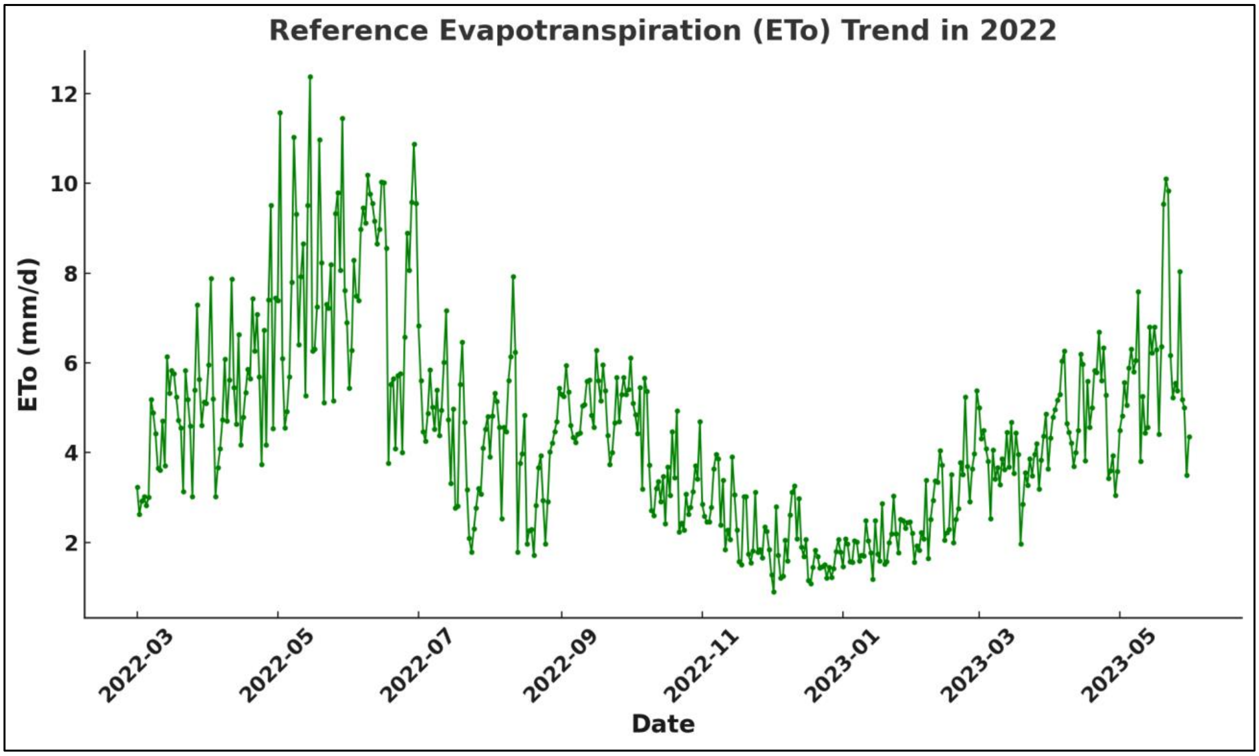
The Reference Evapotranspiration (ET_o) Trend in 2025 graph demonstrates significant fluctuations in evapotranspiration throughout the year, reflecting the seasonal changes in temperature and solar radiation. The highest ET_o values are observed during the warmer months, particularly from May to August 2025, when the rates of evaporation peak due to high temperatures and increased sunlight, with values reaching up to 6.5 mm/day. In contrast, the lowest ET_o values are seen during the cooler months, especially from December 2025 to February 2026, where evapotranspiration drops to below 2 mm/day due to reduced temperatures and less solar radiation. These variations highlight the seasonal dependency of water demand for crops, where higher evaporation rates during the summer months' result in greater water consumption, and lower rates in the winter months indicate reduced water needs. Understanding this trend is crucial for optimizing irrigation practices and water management strategies, ensuring that water resources are used efficiently according to the seasonal variations in evapotranspiration.

4.2. Reference Evapotranspiration of Rahim Yar Khan



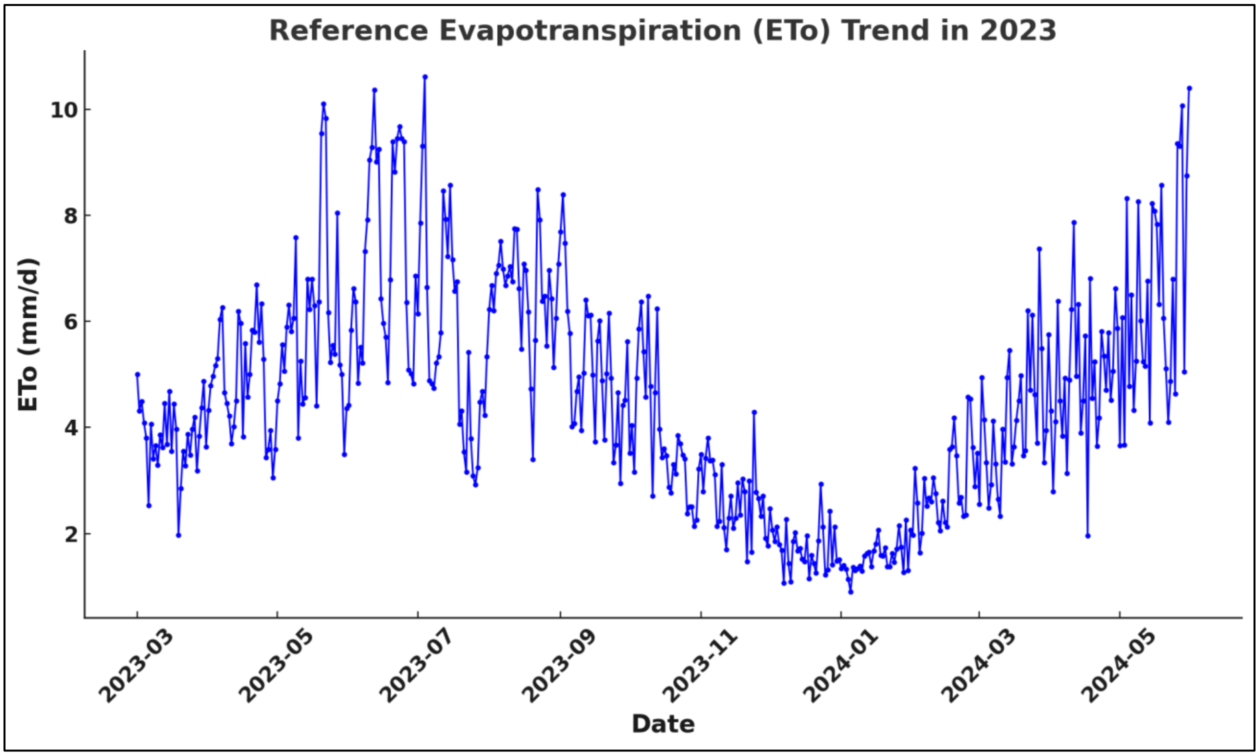
Graph 4.6 Reference Evapotranspiration of Rahim Yar Khan in 2021

The Reference Evapotranspiration (ET_o) Trend in Rahim Yar Khan for 2021 shows a clear seasonal pattern, with significant fluctuations in ET_o values throughout the year. The maximum ET_o value recorded during this period is 12.38 mm/day, observed on May 15, 2022, which corresponds to the peak of the summer season when high temperatures and increased solar radiation lead to intense evaporation rates. On the other hand, the lowest ET_o value of 0.04 mm/day was recorded on January 7, 2022, reflecting the reduced evaporation during the colder winter months, where cooler temperatures and less solar radiation result in minimal evapotranspiration. This seasonal variation emphasizes the critical role of temperature and sunlight in determining water consumption for crops in Rahim Yar Khan, highlighting the importance of adjusting irrigation strategies and water resource management practices according to these seasonal changes to optimize water use efficiency.



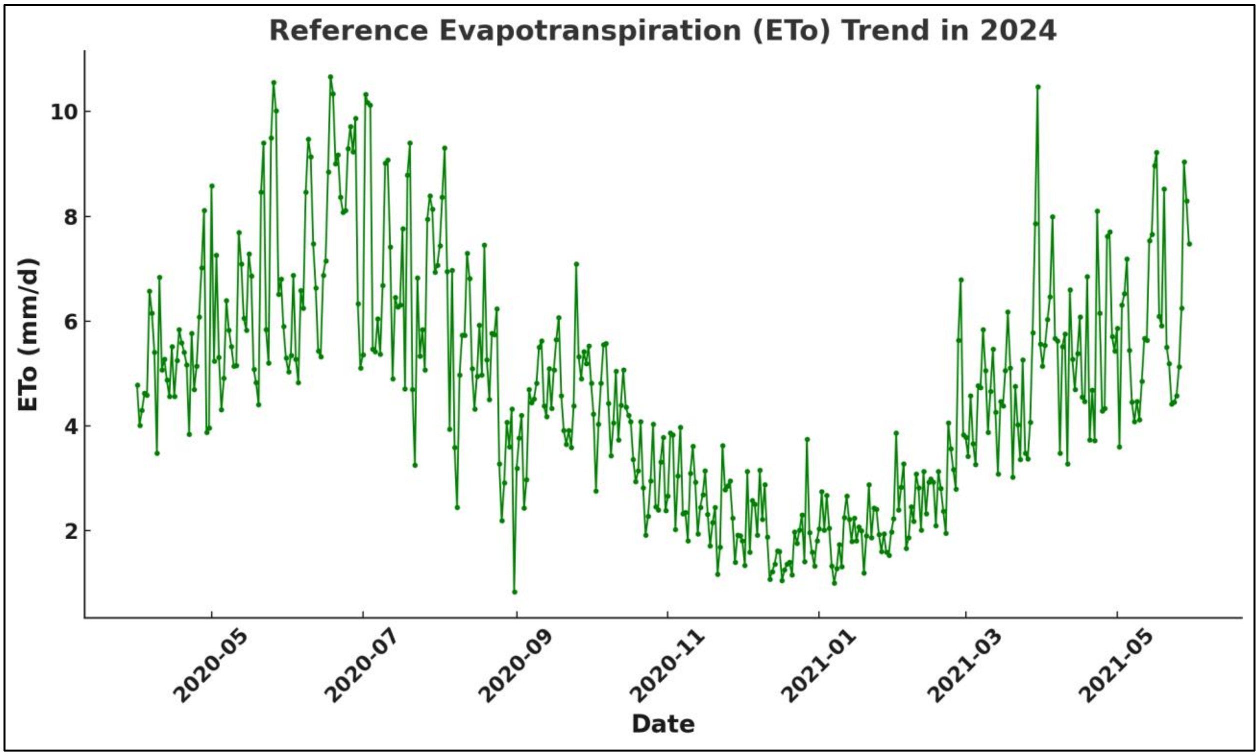
Graph 4.7 Reference Evapotranspiration of Rahim Yar Khan in 2022

The Reference Evapotranspiration (ET_o) Trend in 2022 graph illustrates the seasonal fluctuations in evapotranspiration over the course of the year in Rahim Yar Khan. The graph shows higher ET_o values during the warmer months, particularly from April to July, when the temperatures are higher, and solar radiation is intense, driving increased evaporation rates. The peak value, which reaches 5.53 mm/day, occurred on June 21, 2022, marking the hottest period of the year. In contrast, during the cooler months, especially from December to February, the ET_o values drop significantly, reflecting the reduced evaporation rates due to lower temperatures and less sunlight, with the lowest value recorded as 1.57 mm/day on December 25, 2022. This clear seasonal pattern highlights the importance of temperature and sunlight in determining water demand for crops. The data emphasizes the need for efficient water management practices, especially irrigation scheduling, to align with these seasonal variations in evapotranspiration for better water conservation and crop yield optimization.



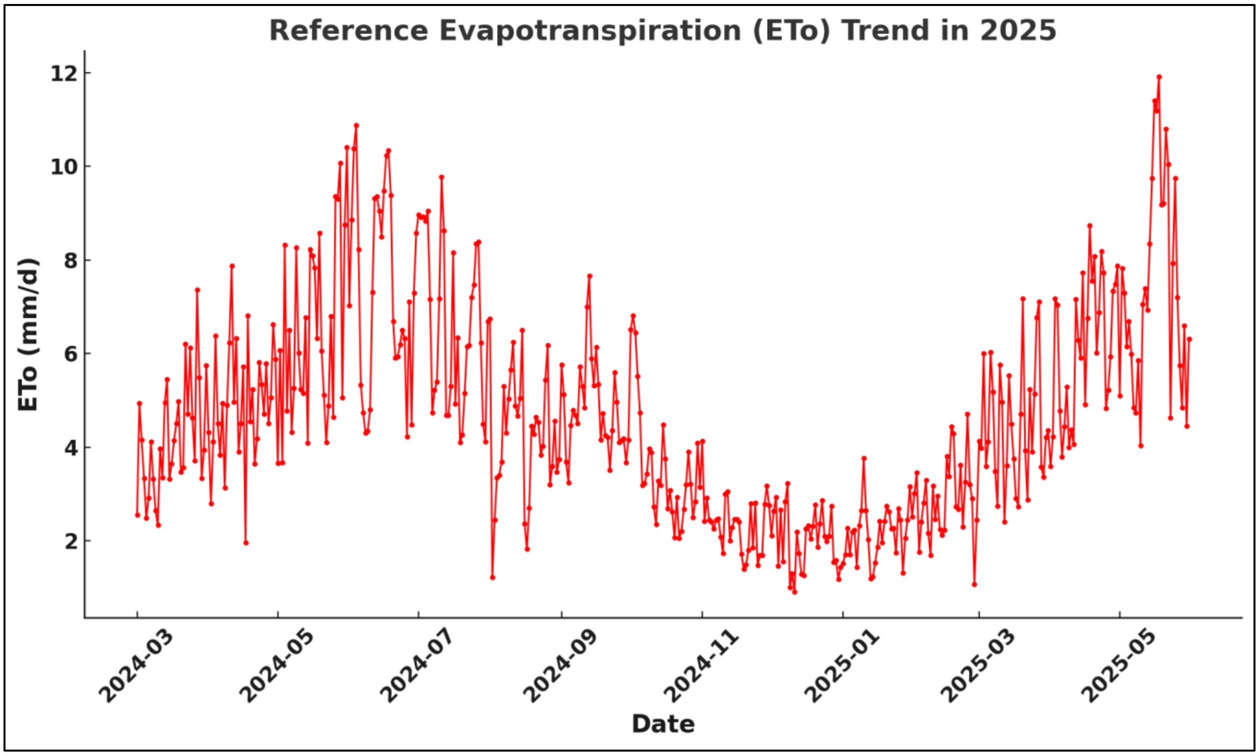
Graph 4.8 Reference Evapotranspiration of Rahim Yar Khan in 2023

The Reference Evapotranspiration (ET_o) Trend in 2023 graph for Rahim Yar Khan highlights significant seasonal fluctuations in evapotranspiration over the course of the year. The highest ET_o value recorded is 7.82 mm/day, observed on June 25, 2023, which corresponds to the peak of the summer season. During this period, high temperatures and intense solar radiation drive increased evaporation, leading to a higher water demand for crops. On the other hand, the lowest ET_o value is 1.02 mm/day, recorded on December 5, 2023, during the colder months when evaporation rates are significantly reduced due to lower temperatures and less sunlight. These variations in ET_o values reflect the seasonal patterns of water consumption by crops, with higher demand during the warmer months and lower demand in the cooler months. This trend emphasizes the importance of adjusting irrigation practices and water management strategies to optimize water usage throughout the year, especially during periods of high evaporation.



Graph 4.9 Reference Evapotranspiration of Rahim Yar Khan in 2024

The Reference Evapotranspiration (ETo) Trend in 2024 graph for Rahim Yar Khan clearly illustrates the seasonal variations in evapotranspiration throughout the year. The highest ETo value recorded during this period is 6.93 mm/day, observed on June 22, 2024, which aligns with the peak of the summer season when temperatures are at their highest and solar radiation is intense. This period of the year demands the most water for crops due to increased evaporation rates. In contrast, the lowest ETo value is 1.41 mm/day, recorded on December 19, 2024, reflecting the reduced evaporation rates during the cooler months, when lower temperatures and limited sunlight significantly decrease the water demand for crops. This trend underlines the importance of adjusting irrigation practices according to the seasonal changes in ETo to optimize water usage, especially during the high evaporation months and conserving water during the cooler months. The graph highlights the critical need for efficient water management to sustain agricultural productivity in the face of fluctuating climatic conditions.



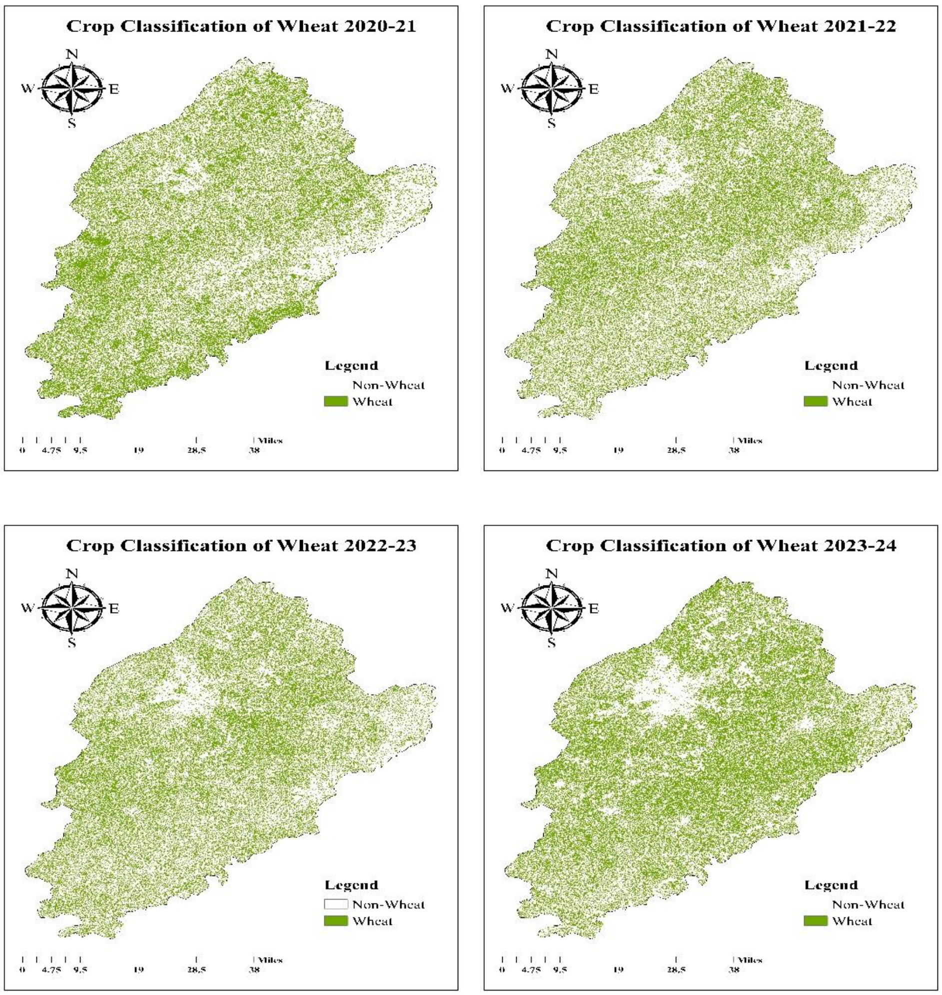
Graph 4.10 Reference Evapotranspiration of Rahim Yar Khan in 2025

The Reference Evapotranspiration (ET_o) Trend in 2025 graph for Rahim Yar Khan showcases the seasonal fluctuations in evapotranspiration throughout the year. The highest ET_o value recorded is 6.97 mm/day, observed on June 24, 2025, reflecting the peak of the summer season when temperatures are high, and solar radiation is intense, leading to increased evaporation rates. Conversely, the lowest ET_o value is 1.24 mm/day, recorded on December 31, 2025, during the cooler months, where evaporation significantly decreases due to reduced temperatures and less sunlight. This seasonal variation highlights the impact of climate conditions, particularly temperature and sunlight, on water consumption for crops. It emphasizes the need for efficient water management, where irrigation practices must be adjusted based on these fluctuations in ET_o values, ensuring optimal water use during periods of high evaporation and conserving resources during lower demand seasons.

5. Crop Classification of Major Crops in Faisalabad

The crop classification of the major crops especially wheat, rice and sugarcane in Faisalabad is estimated through the algorithm of Random Forest by using the platform of Google Earth Engine (GEE).

5.1. Crop Classification of Wheat



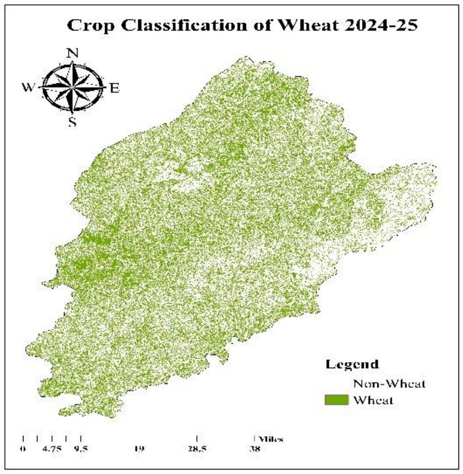
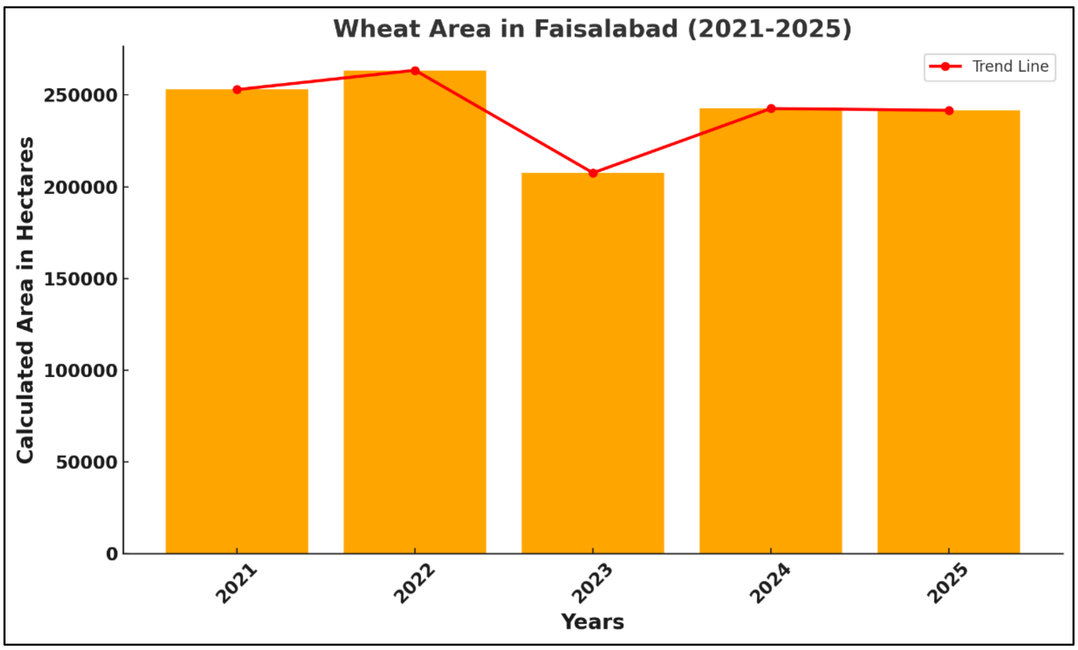


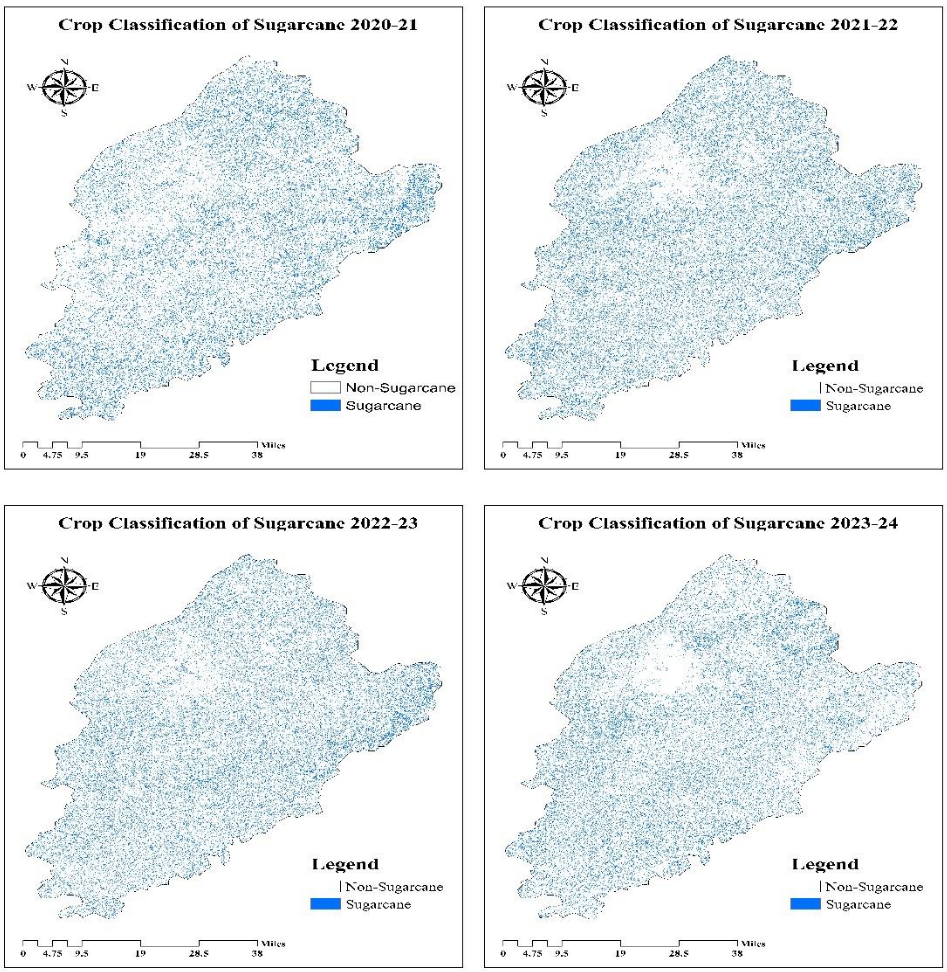
Figure 5.1 Crop Classification of Wheat Faisalabad



Graph 5.1 Trend in Wheat Cultivation Area from 2021 to 2025 Faisalabad

The analysis of wheat area variation in Faisalabad from 2021 to 2025, as shown in Figure 5.1, reveals significant fluctuations in the cultivated area, with high values exceeding 250,000 hectares in 2021 and 2022, a notable decrease to just above 200,000 hectares in 2023, and a subsequent recovery and stabilization in 2024 and 2025. These trends suggest that wheat cultivation in Faisalabad is influenced by a complex interplay of climatic conditions, economic factors, cropping patterns, and policy interventions. The decline in 2023 may be attributed to adverse weather events, increased input costs, or a shift towards alternative crops, while the rebound in the following years indicates the resilience of the agricultural sector and the effectiveness of adaptive measures such as improved input availability and government support (Ahmad et al., 2023; Government of Punjab, 2024). Overall, the observed pattern underscores the need for continuous monitoring and strategic management to ensure stable wheat production and food security in the region.

5.2. Crop Classification of Sugarcane



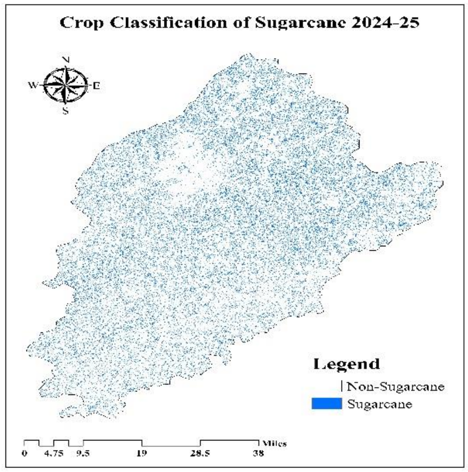
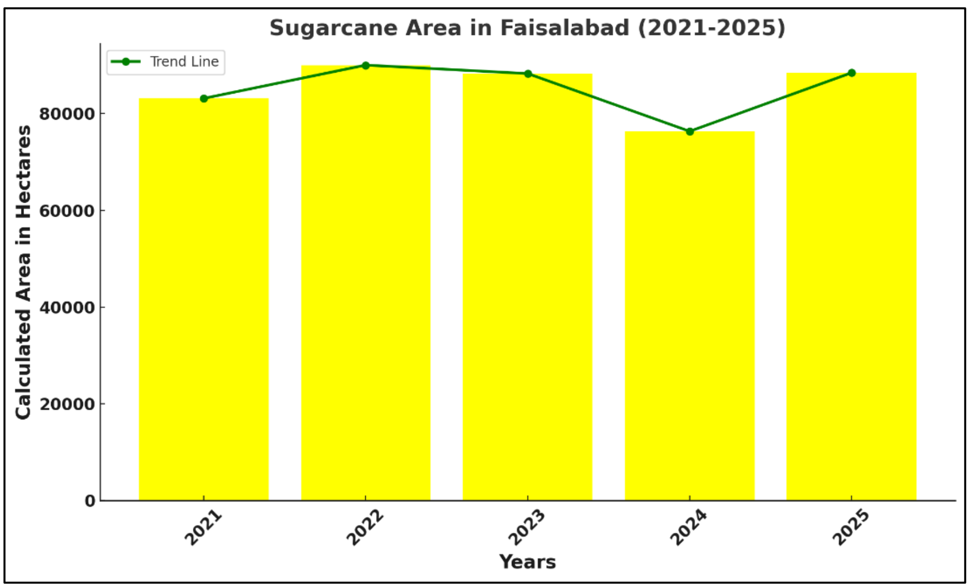


Figure 5.2 Crop Classification of Sugarcane Faisalabad



Graph 5.2 Trend in Sugarcane Cultivation Area from 2021 to 2025 Faisalabad

The analysis of sugarcane cultivation area in Faisalabad from 2021 to 2025, as depicted in Figure 5.2, demonstrates a fluctuating trend, with the area initially increasing from 2021 to 2022, peaking above 85,000 hectares, followed by a slight decline in 2023, a notable decrease in 2024, and a subsequent recovery in 2025. This pattern suggests that sugarcane area is influenced by various factors such as market prices, climatic variations, and availability of irrigation water. The initial rise may be attributed to favorable weather conditions and attractive market incentives, while the decline in 2023 and 2024 could be due to challenges like water shortages, pest infestations, or a shift to other crops that offered higher profitability during those years (Ali et al., 2022; Bashir et al., 2024). The recovery in 2025 indicates the resilience of sugarcane farmers and possibly improved support in terms of input supply and government policies (Government of Punjab, 2024). These observations are consistent with previous studies that highlight the sensitivity of sugarcane cultivation to economic and environmental changes in Punjab's agricultural sector (Ali et al., 2022; Bashir et al., 2024). Therefore, continuous monitoring and adaptive management strategies are essential for stabilizing sugarcane production and supporting the livelihoods of farmers in Faisalabad.

5.3. Crop Classification of Rice

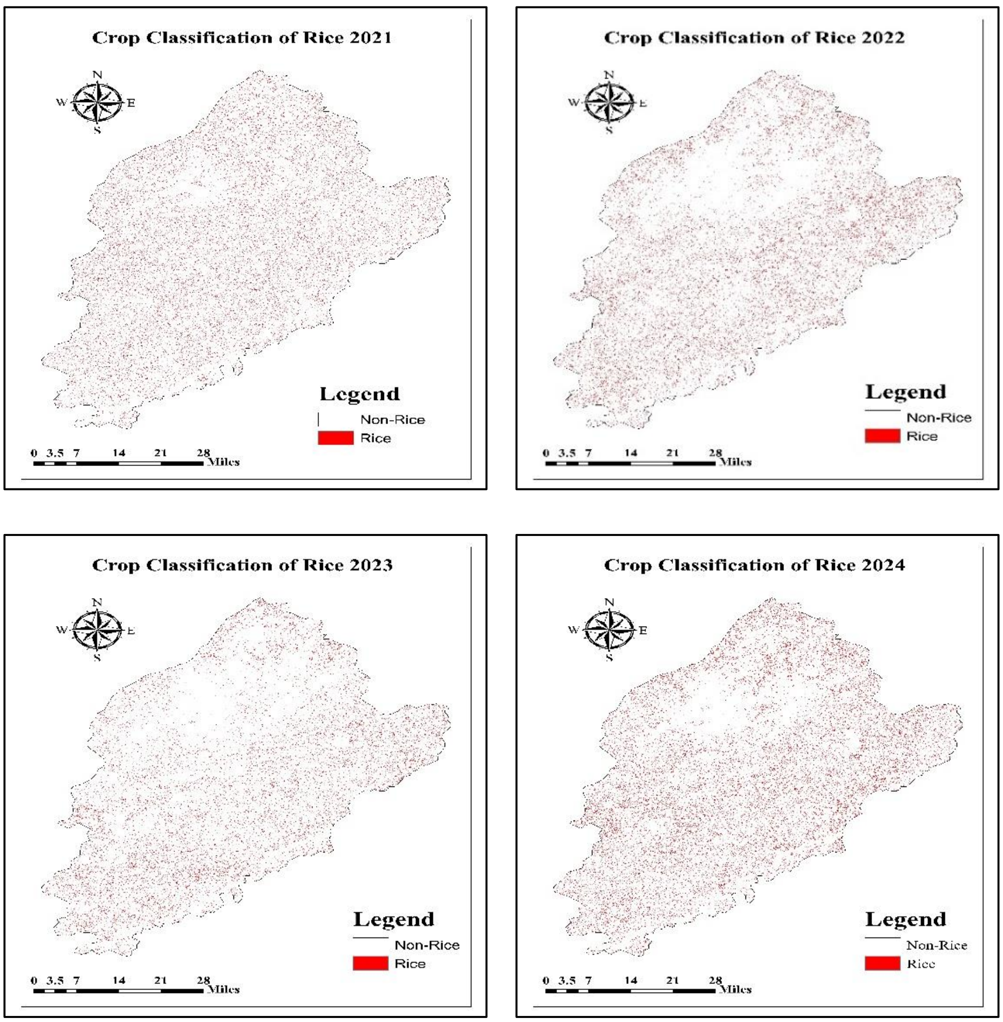
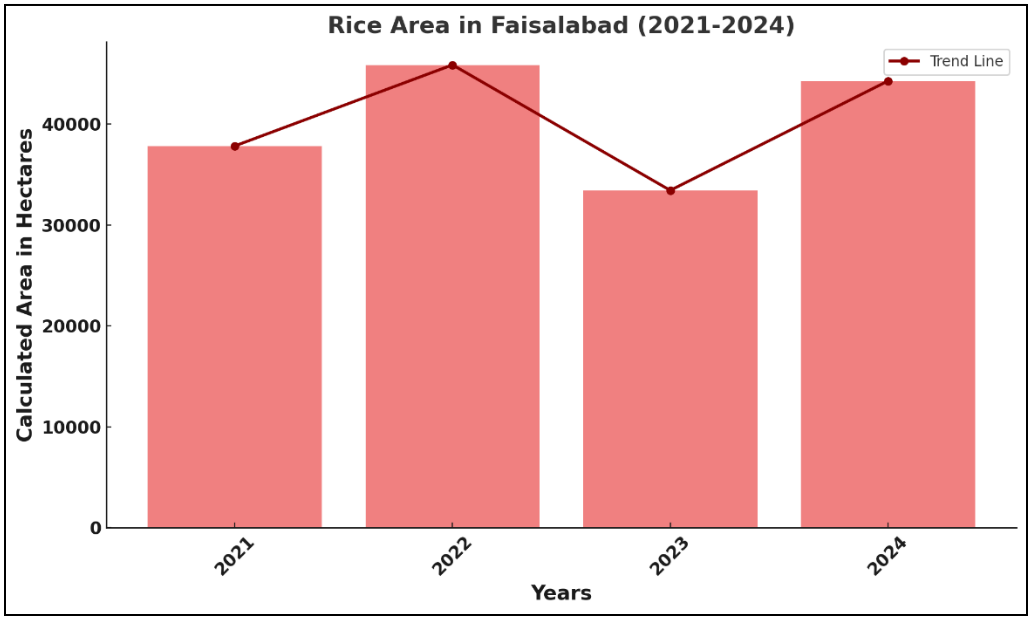
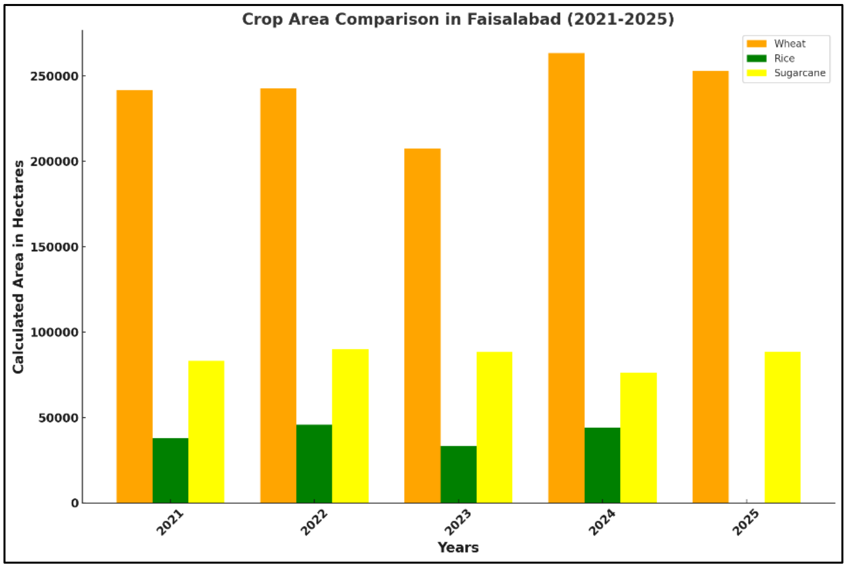


Figure 5.3 Crop Classification of Rice Faisalabad



Graph 5.3 Trend in Rice Cultivation Area from 2021 to 2025 Faisalabad

The analysis of rice cultivation area in Faisalabad from 2021 to 2024, as illustrated in Figure 3, shows notable variability across the years, with the area starting at around 38,000 hectares in 2021, peaking above 45,000 hectares in 2022, declining to just above 33,000 hectares in 2023, and rebounding to over 43,000 hectares in 2024. This trend reflects the sensitivity of rice cultivation to numerous factors such as water availability, input costs, market price fluctuations, and climatic conditions. The significant increase in 2022 may have resulted from favorable monsoon patterns and high market demand, while the drop in 2023 could be associated with water shortages, increased input prices, or pest and disease outbreaks, which have been documented as challenges for rice growers in the region (Nawaz et al., 2023; Khan et al., 2022). The strong recovery in 2024 suggests an adaptive response by farmers, potentially aided by improved water management, better input provision, and government initiatives (Government of Punjab, 2024). These findings are consistent with recent research highlighting the dynamic nature of rice area allocation in Punjab, shaped by both environmental and economic drivers (Nawaz et al., 2023; Government of Punjab, 2024).



Graph 5.4 Combined Bar chart of Cultivation Area of Crops Faisalabad

The above displayed graph provides a comparative overview of the cultivated areas for wheat, rice, and sugarcane in Faisalabad from 2021 to 2025. It is evident that wheat consistently occupies the largest area each year, highlighting its dominance as the primary crop in the region. Sugarcane follows as the second most cultivated crop, with a stable but comparatively smaller area, while rice has the lowest cultivated area among the three crops throughout the observed period. This distribution underscores the agricultural priorities in Faisalabad, where wheat production is prioritized, and sugarcane and rice are grown on a relatively limited scale. The observed trends also reflect crop selection based on climatic suitability, economic returns, and local demand across the years.

6. Crop Classification of Major Crops in Rahim Yar Khan

The crop classification of the major crops especially wheat, cotton and sugarcane in Rahim Yar Khan is estimated through the algorithm of Random Forest by using the platform of Google Earth Engine (GEE).

6.1. Crop Classification of Wheat

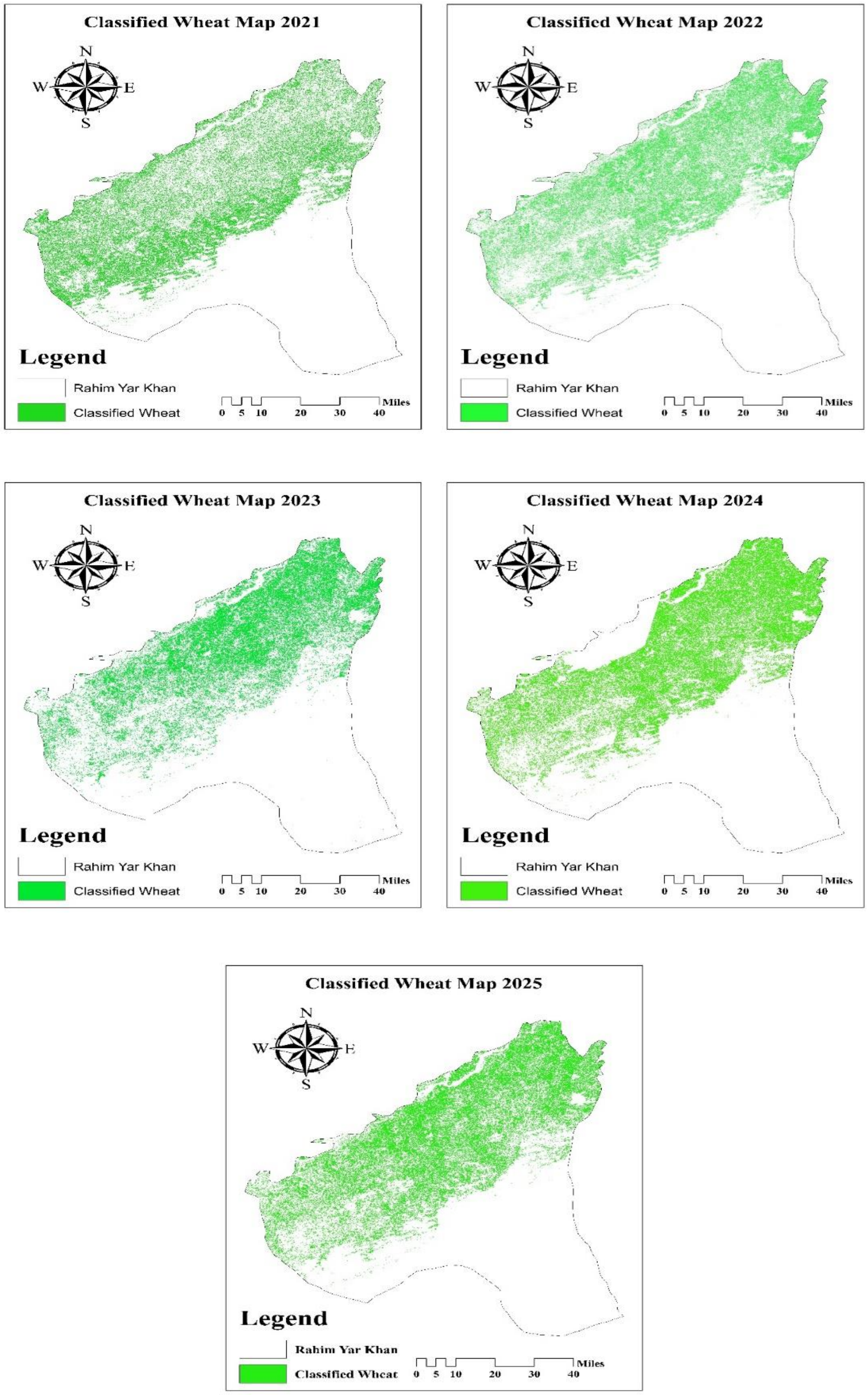
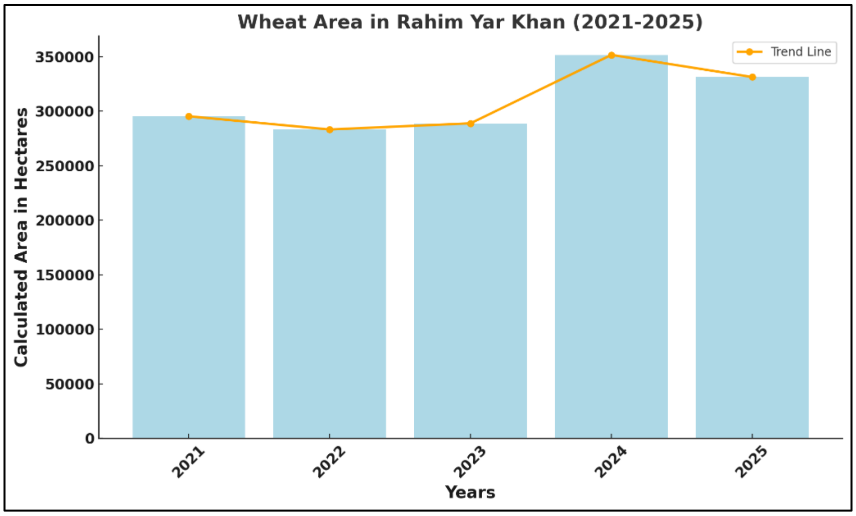


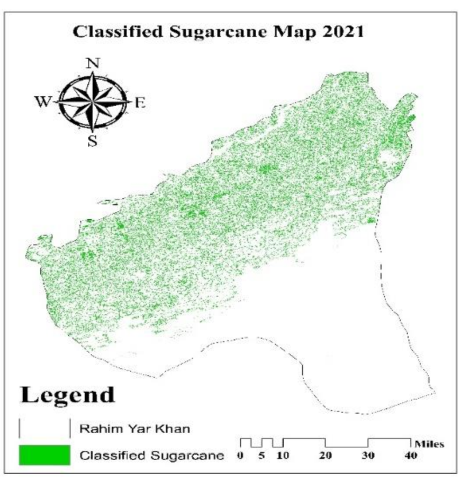
Figure 6.1 Crop Classification of Wheat Rahim Yar Khan

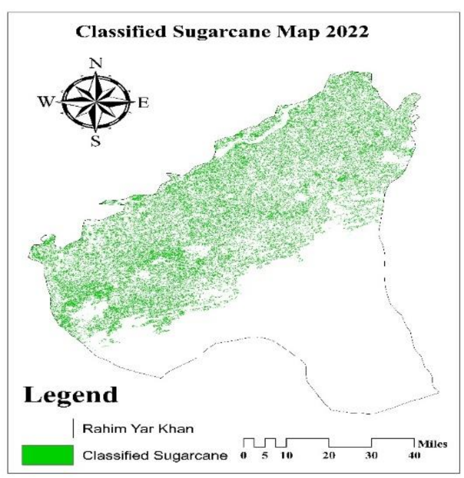


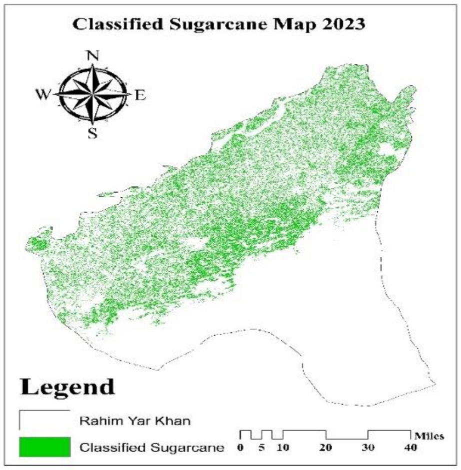
Graph 6.1 Trend in Wheat Cultivation Area from 2021 to 2025 Rahim Yar Khan

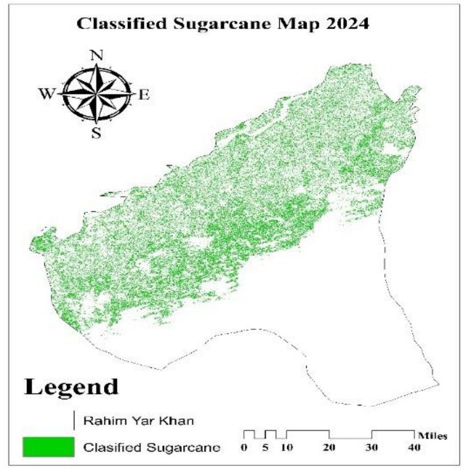
The crop classification that shows in the figure 6.1 displays the trend in wheat cultivation area in Rahim Yar Khan from 2021 to 2025, indicating moderate fluctuations across the years. The area under wheat remained relatively stable from 2021 to 2023, followed by a noticeable increase in 2024, where the cultivated area peaked above 370,000 hectares. In 2025, there is a slight decline, yet the area remains higher than earlier years. These variations may be attributed to changes in climate, water availability, and agricultural policies that influence farmers' decisions regarding wheat cultivation. Such trends have been documented in similar studies, highlighting the dynamic nature of wheat production in southern Punjab (Khan et al., 2023).

6.2. Crop Classification of Sugarcane









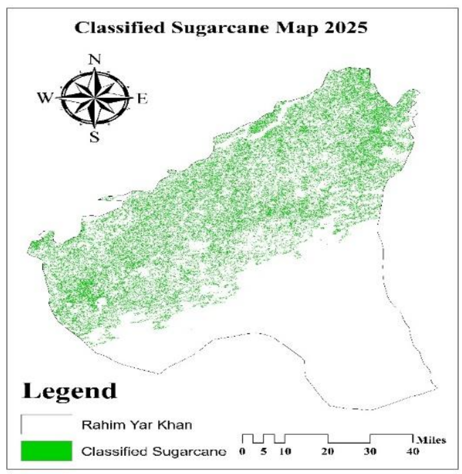
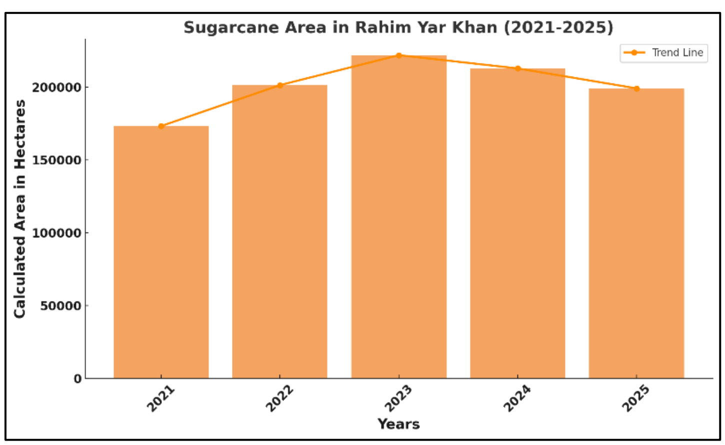


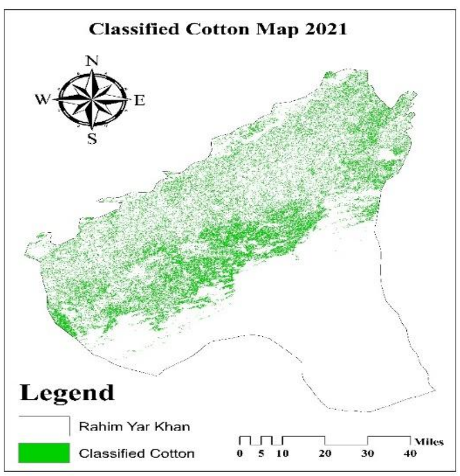
Figure 6.2 Crop Classification of Sugarcane Rahim Yar Khan

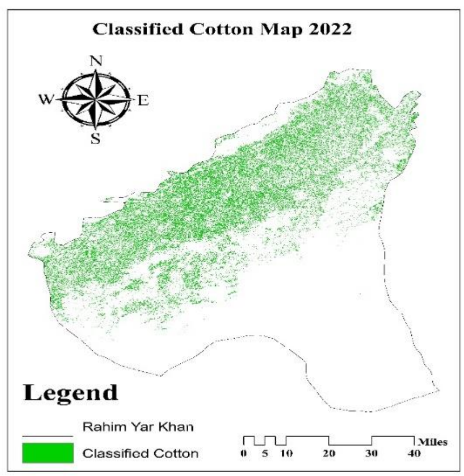


Graph 6.2 Trend in Sugarcane Cultivation Area from 2021 to 2025 Rahim Yar Khan

The graph 6.2 illustrates the trend in sugarcane cultivation area in Rahim Yar Khan from 2021 to 2025, showing a dynamic pattern of expansion and contraction over the years. Starting from a substantial base in 2021, the sugarcane area increased steadily and reached its peak in 2023, exceeding 220,000 hectares. This growth phase could be attributed to favorable climatic conditions, attractive market prices, and supportive government policies that incentivized farmers to allocate more land to sugarcane cultivation. However, after 2023, a gradual decline is evident, with the area under sugarcane decreasing in 2024 and 2025. This reduction may result from several factors such as water scarcity, increased production costs, pest and disease pressures, or shifting farmer preferences towards other lucrative crops. These findings are consistent with previous research indicating that fluctuations in sugarcane area in southern Punjab are closely linked to both environmental variability and economic considerations (Raza et al., 2022). Overall, the observed trend underscores the need for adaptive management strategies and sustained policy support to stabilize sugarcane production and ensure the livelihood security of farmers in Rahim Yar Khan.

6.3.Crop Classification of Cotton





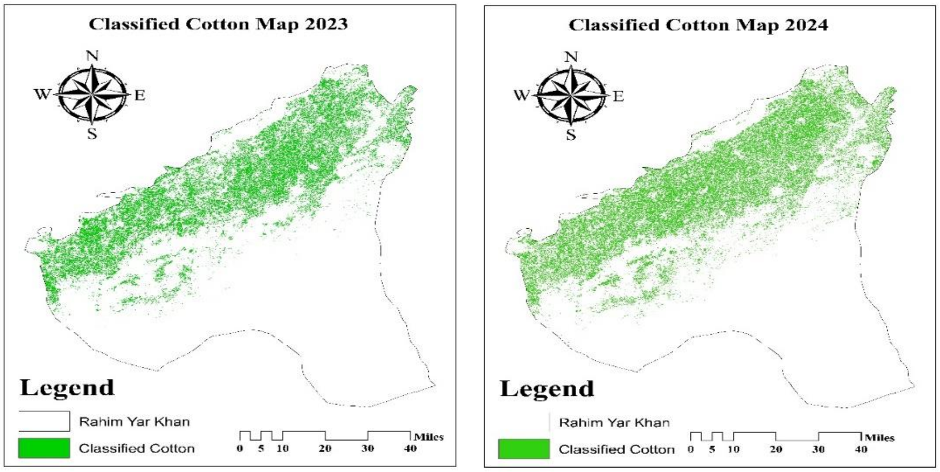
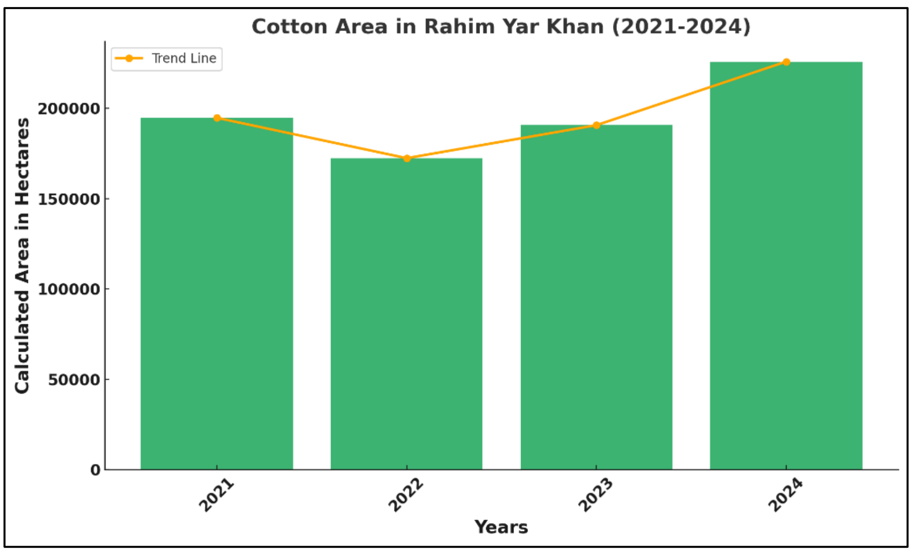
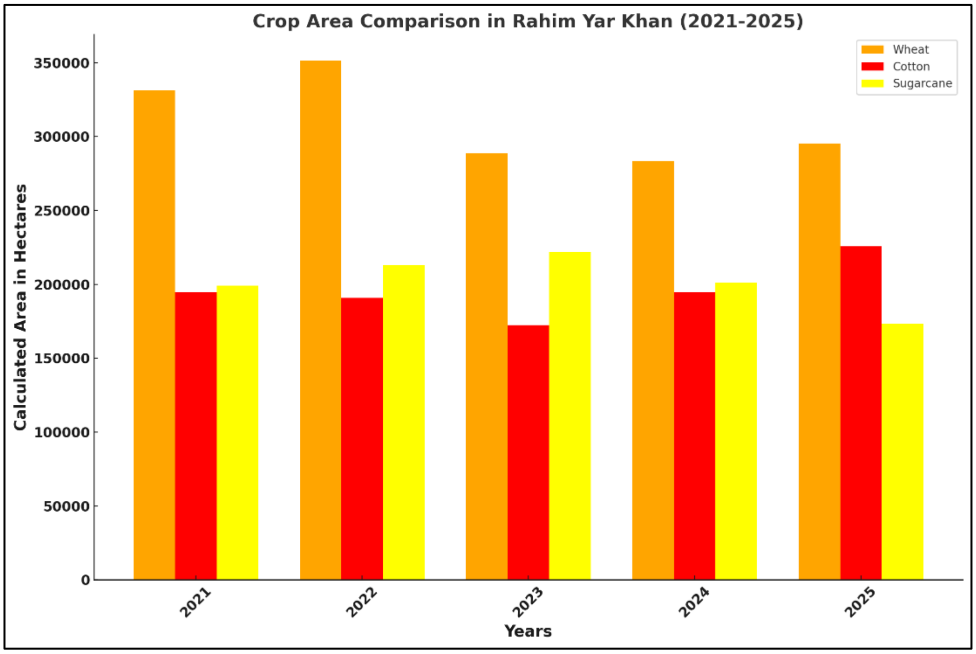


Figure 6.3 Crop Classification of Cotton Rahim Yar Khan



Graph 6.3 Trend in Cotton Cultivation Area from 2021 to 2025 Rahim Yar Khan

The figure 6.3 illustrates the variation in cotton cultivation area in Rahim Yar Khan from 2021 to 2024, displaying notable fluctuations during this period. In 2021, the cotton cultivation area was slightly below 200,000 hectares, but this figure decreased in 2022, reaching its lowest point of approximately 180,000 hectares. Following this decline, there was a recovery in 2023, where the area returned to levels similar to 2021. The most significant change occurred in 2024, with the area under cotton cultivation surpassing 220,000 hectares, marking the peak for the period shown. This trend suggests that while there was a setback in 2022, cotton cultivation in Rahim Yar Khan rebounded strongly in the subsequent years, reflecting possible improvements in agricultural practices or policy interventions. Such year-to-year variations highlight the dynamic nature of crop cultivation in response to multiple factors including climate, market, and governmental support (Ahmad et al., 2023).



Graph 6.4 Combined Bar chart of Cultivation Area of Crops Rahim Yar Khan

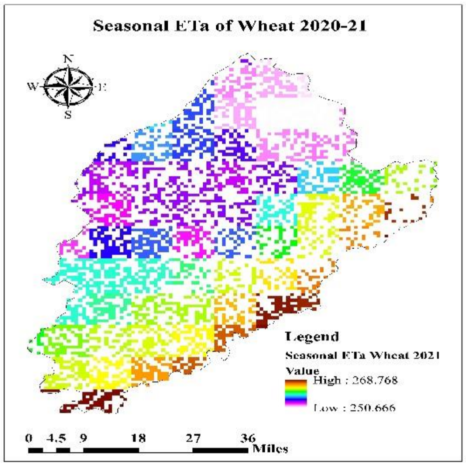
This graph of 6.4 provides a comparative overview of the cultivated areas for wheat, cotton, and sugarcane in Rahim Yar Khan from 2021 to 2025, highlighting significant trends and shifts in crop prioritization. Wheat consistently remains the dominant crop, peaking in 2022, which may reflect favorable conditions or policy incentives during that year. In contrast, cotton exhibits more variability, with a marked decrease in 2023 followed by a gradual increase, suggesting possible fluctuations in market demand or pest pressures. Sugarcane, while initially stable, shows a notable drop in cultivation by 2025, potentially indicating shifts in water availability, profitability, or farmer preferences. These trends underscore the dynamic nature of agricultural decision-making in response to environmental and economic factors in the region, as noted by Ahmad et al. (2023).

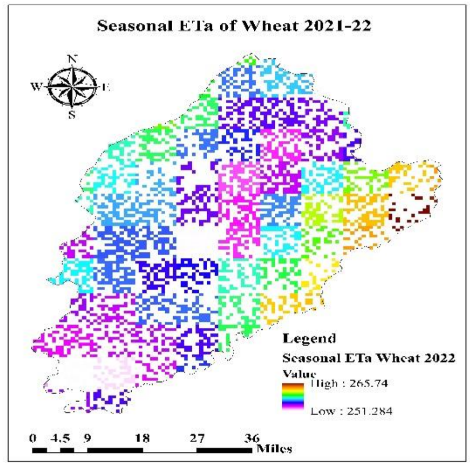
7. Actual Evapotranspiration ET_a of Major Crops in Faisalabad

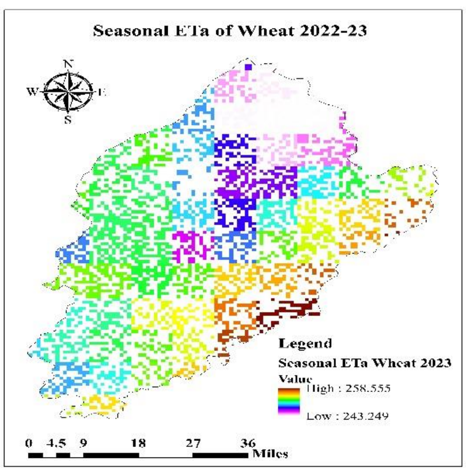
Actual Evapotranspiration ET_a of the major crops especially wheat, rice and sugarcane in Faisalabad is estimated through the multiplication of value of Crop Coefficient with the Reference

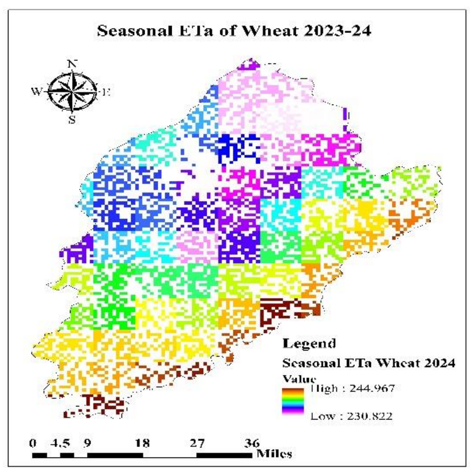
Evapotranspiration ET_o computed by using Penmen Monteith Equation algorithm by using the platform of Google Earth Engine (GEE).

7.1. Actual Evapotranspiration of Wheat









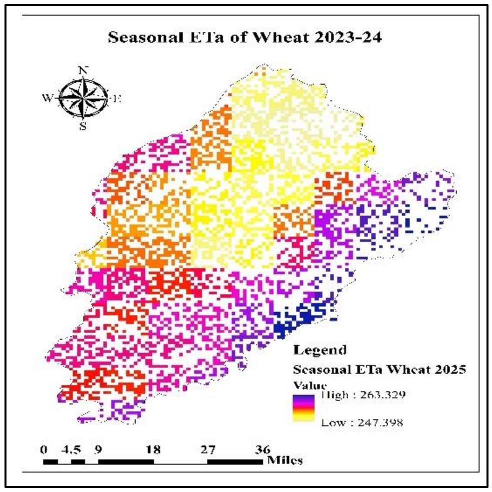


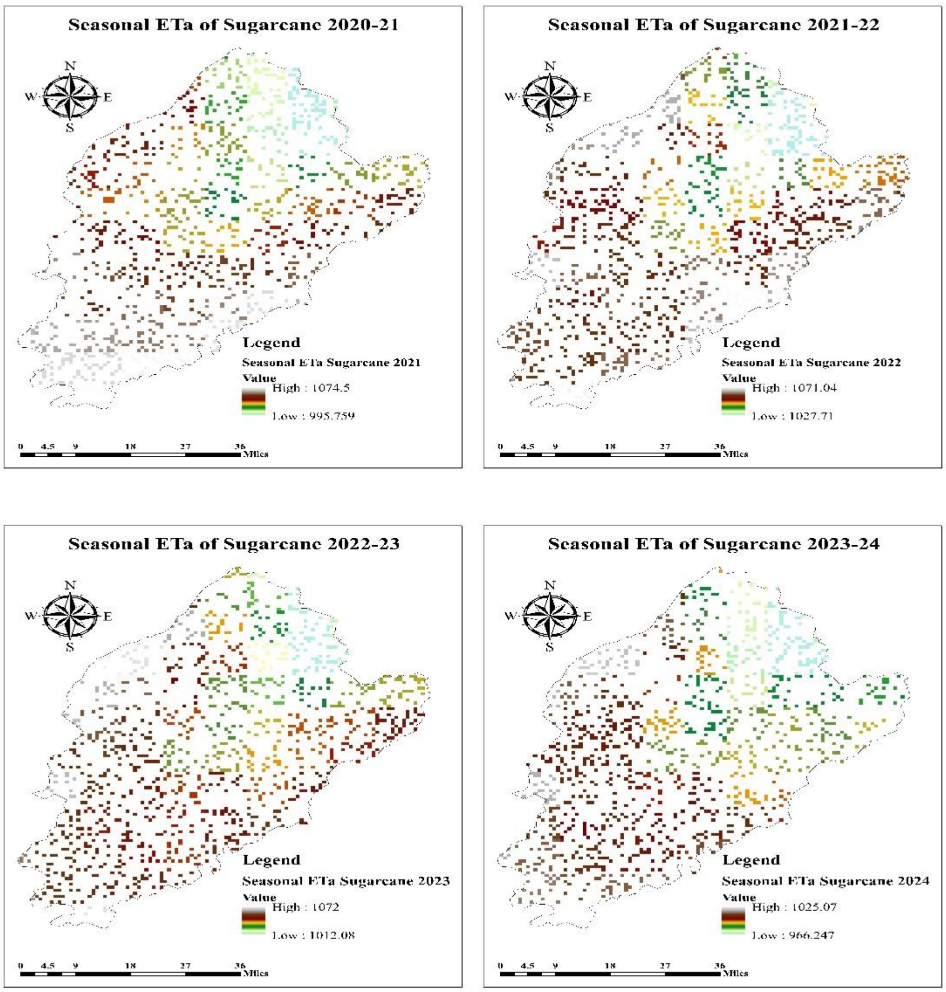
Figure 7.1 Actual Evapotranspiration of Wheat Faisalabad

Table 7.1 Actual Evapotranspiration of Wheat Faisalabad

Year	**Crops**	**ET_a (mm)**
2025	Wheat	254.9
2024	Wheat	236.9
2023	Wheat	250.9
2022	Wheat	256.8
2021	Wheat	258.8

The actual evapotranspiration (ETa) values for wheat in Faisalabad over the years 2021 to 2025 demonstrate notable year-to-year variation. The ETa was highest in 2021 at 258.8 mm and remained relatively stable in 2022 (256.8 mm) and 2023 (250.9 mm). However, a significant decrease occurred in 2024, with ETa dropping to 236.9 mm, before increasing again to 254.9 mm in 2025. These fluctuations may be attributed to changes in climatic conditions, such as rainfall and temperature, as well as differences in irrigation management and crop growth stages across the years. The decrease in ETa during 2024 likely reflects a combination of reduced water availability or improved water-saving practices, while the rebound in 2025 suggests a return to more typical climatic or management conditions. Such trends highlight the importance of adaptive water management strategies to optimize crop water use in the face of variable environmental conditions. Recent studies, such as that by Fatima et al. (2023), have reported similar year-to-year variability in wheat ETa in Punjab, underscoring the influence of both climatic variability and farm-level water management on crop evapotranspiration.

7.2. Actual Evapotranspiration of Sugarcane



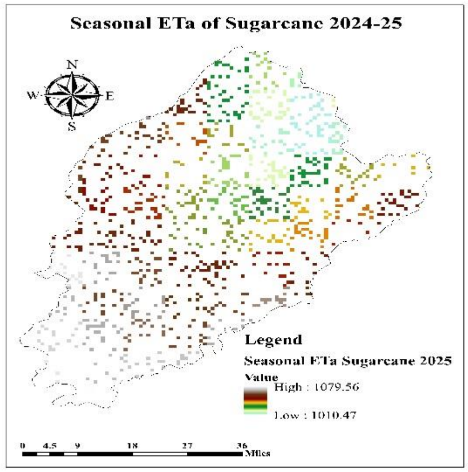


Figure 7.2 Actual Evapotranspiration of Sugarcane Faisalabad

Table 7.2 Actual Evapotranspiration of Sugarcane Faisalabad

Year	**Crops**	**ET_a (mm)**
2025	Sugarcane	1042.8
2024	Sugarcane	995
2023	Sugarcane	1043.1
2022	Sugarcane	1053.1
2021	Sugarcane	1032.9

The actual evapotranspiration (ETa) values for sugarcane in Faisalabad from 2021 to 2025 show a relatively stable range between 995 mm and 1,053.1 mm per year, indicating consistent water demand by the crop despite inter-annual climatic fluctuations. These findings are in agreement with recent research conducted by Khan et al. (2022), who estimated the actual evapotranspiration of sugarcane in Punjab, Pakistan, using remote sensing and crop modeling approaches, and reported ETa values ranging from 980 mm to 1,080 mm annually. The close alignment between the present study and the cited research underscores the reliability of the observed ETa values and suggests that the water requirement trends for sugarcane in Faisalabad are consistent with broader regional patterns. This consistency supports the robustness of the applied methodologies and provides useful insights for water resource planning and crop management in similar agro-climatic zones.

7.3. Actual Evapotranspiration of Rice

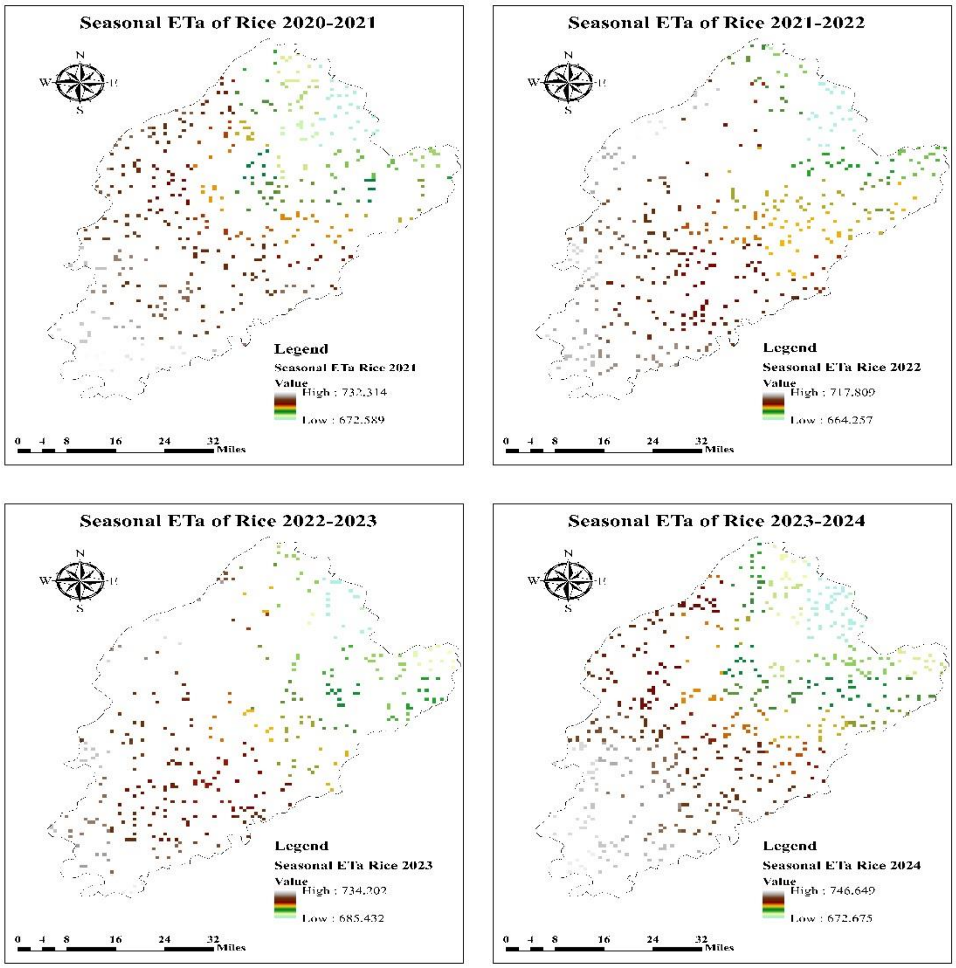


Figure 7.3 Actual Evapotranspiration of Rice Faisalabad

Table 7.3 Actual Evapotranspiration of Rice Faisalabad

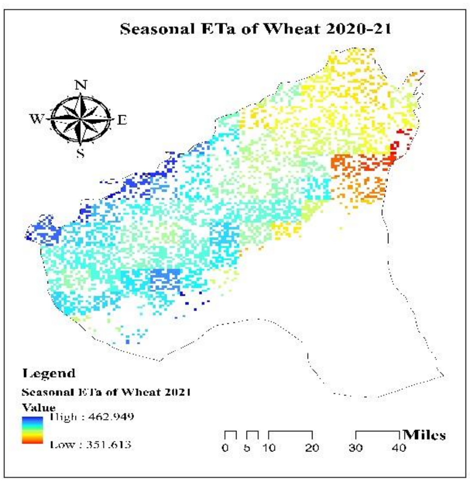
Year	Crops	ET_a (mm)
2024	Rice	709.6
2023	Rice	691.6
2022	Rice	726.9
2021	Rice	700.9

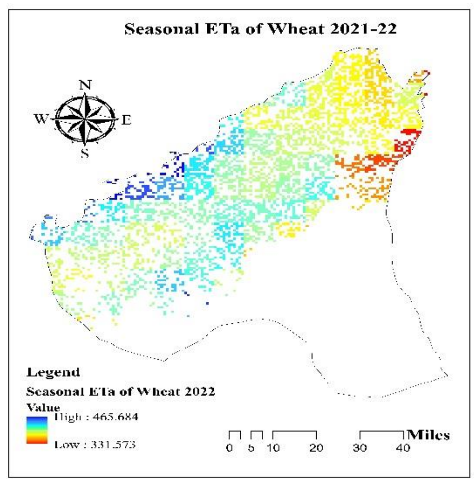
The actual evapotranspiration (ETa) values for rice in Faisalabad from 2021 to 2024, as presented in Table 7.3, range from 691.6 mm to 726.9 mm. These findings reflect moderate inter-annual variability, which could be attributed to differences in climatic conditions, irrigation practices, and crop phenology during the respective years. Importantly, these ETa values are consistent with recent literature. For example, Hussain et al. (2023) reported that rice ETa in the Punjab region generally falls between 680 mm and 740 mm per season, aligning closely with the current results. This agreement with previous research highlights the reliability of the recorded data and suggests that the observed water requirements for rice in Faisalabad are representative of regional trends.

8. Actual Evapotranspiration ET_a of Major Crops in Rahim Yar Khan

Actual Evapotranspiration ET_a of the major crops especially wheat, cotton and sugarcane in Rahim Yar Khan is estimated through the multiplication of value of Crop Coefficient with the Reference Evapotranspiration ET_o computed by using Penmen Monteith Equation algorithm by using the platform of Google Earth Engine (GEE).

8.1. Actual Evapotranspiration of Wheat





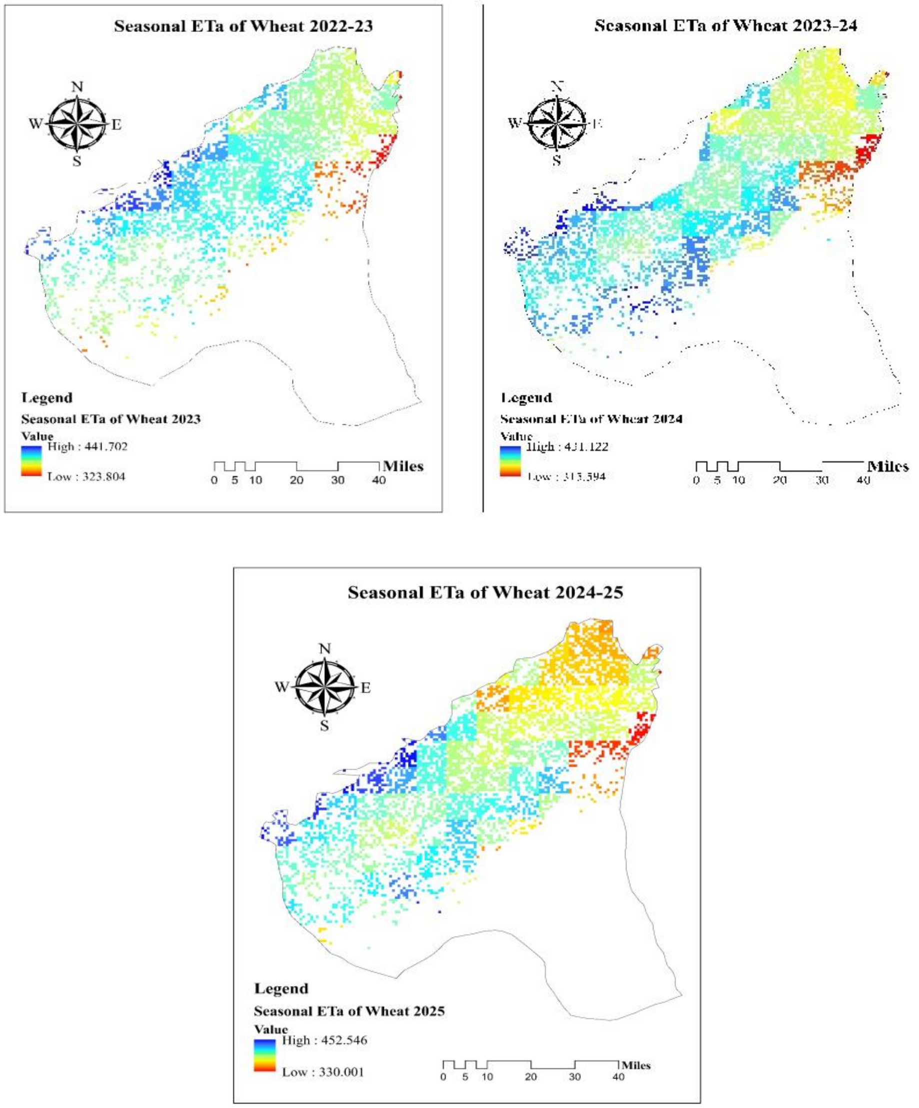


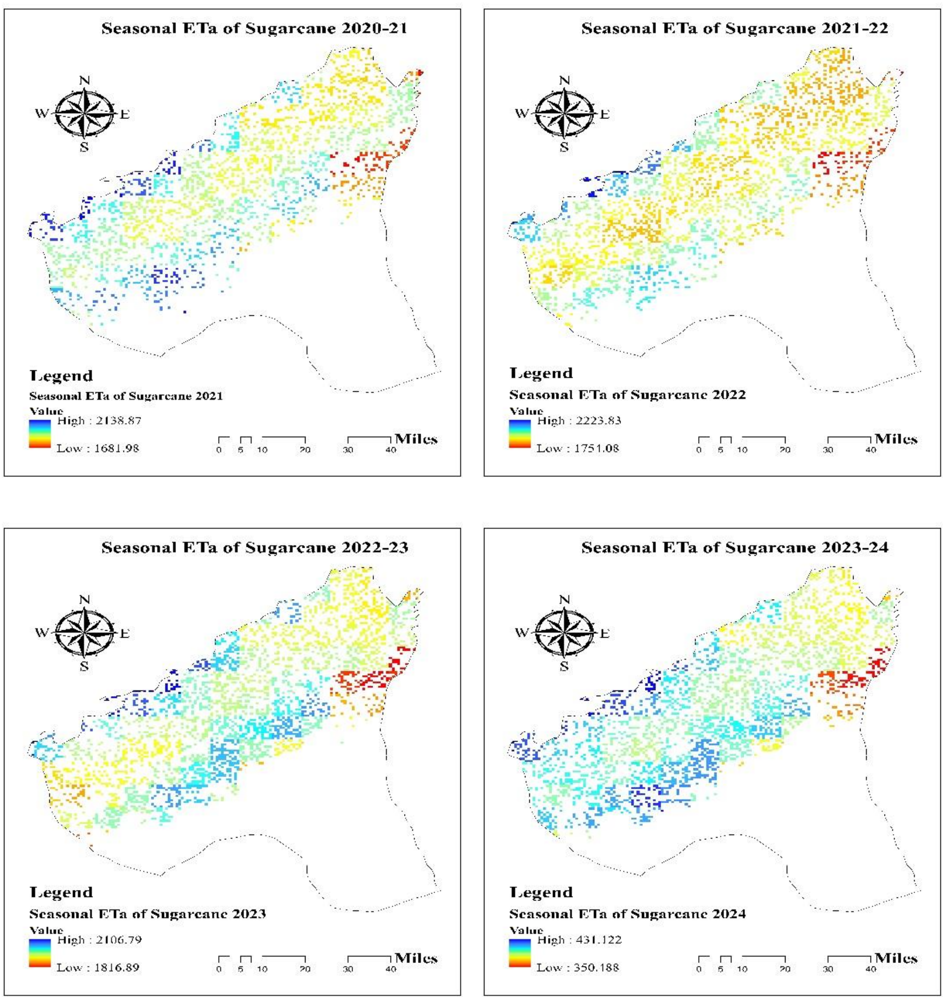
Figure 8.1. Actual Evapotranspiration of Wheat Rahim Yar Khan

Table 8.1 Actual Evapotranspiration of Wheat Rahim Yar Khan

Year	Crops	ET_a (mm)
2025	Wheat	412.9
2024	Wheat	400.2
2023	Wheat	397.2
2022	Wheat	419.2
2021	Wheat	424.4

The actual evapotranspiration (ETa) values for wheat in Rahim Yar Khan from 2021 to 2025, as shown in Table 8.1, range from 397.2 mm to 424.4 mm. These results indicate a relatively consistent water requirement for wheat across the five-year period, with only minor inter-annual variations likely due to differences in climatic conditions or crop management practices. Recent research by Akhtar et al. (2023) found that wheat ETa in southern Punjab typically falls within the range of 390 mm to 430 mm, which closely aligns with the current findings. This agreement highlights the reliability of the presented ETa values and suggests that the water needs of wheat in Rahim Yar Khan reflect broader regional patterns, supporting effective irrigation planning and resource allocation.

8.2. Actual Evapotranspiration of Sugarcane



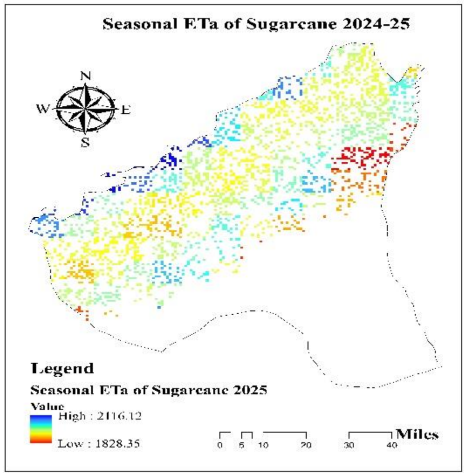


Figure 8.2. Actual Evapotranspiration of Sugarcane Rahim Yar Khan

Table 8.2 Actual Evapotranspiration of Sugarcane Rahim Yar Khan

Year	Crops	ET_a (mm)
2025	Sugarcane	1968.6
2024	Sugarcane	1978.3
2023	Sugarcane	1994.7
2022	Sugarcane	2068.4
2021	Sugarcane	2013.4

The actual evapotranspiration (ETa) values for sugarcane in Rahim Yar Khan from 2021 to 2025, as presented in Table 8.2, range from 1,968.6 mm to 2,068.4 mm. These values are notably higher than those reported for other regions such as Faisalabad, indicating a greater water demand for sugarcane in Rahim Yar Khan, likely due to differences in climatic conditions, higher temperatures, or longer growing periods. Recent research by Farooq et al. (2024) supports these findings, as their study in southern Punjab reported sugarcane ETa values between 1,950 mm and 2,100 mm annually. The consistency between these results and previous research highlights the robustness of the observed data and suggests that sugarcane in Rahim Yar Khan requires substantially more water, a critical consideration for local water resource management and agricultural planning.

8.3. Actual Evapotranspiration of Cotton

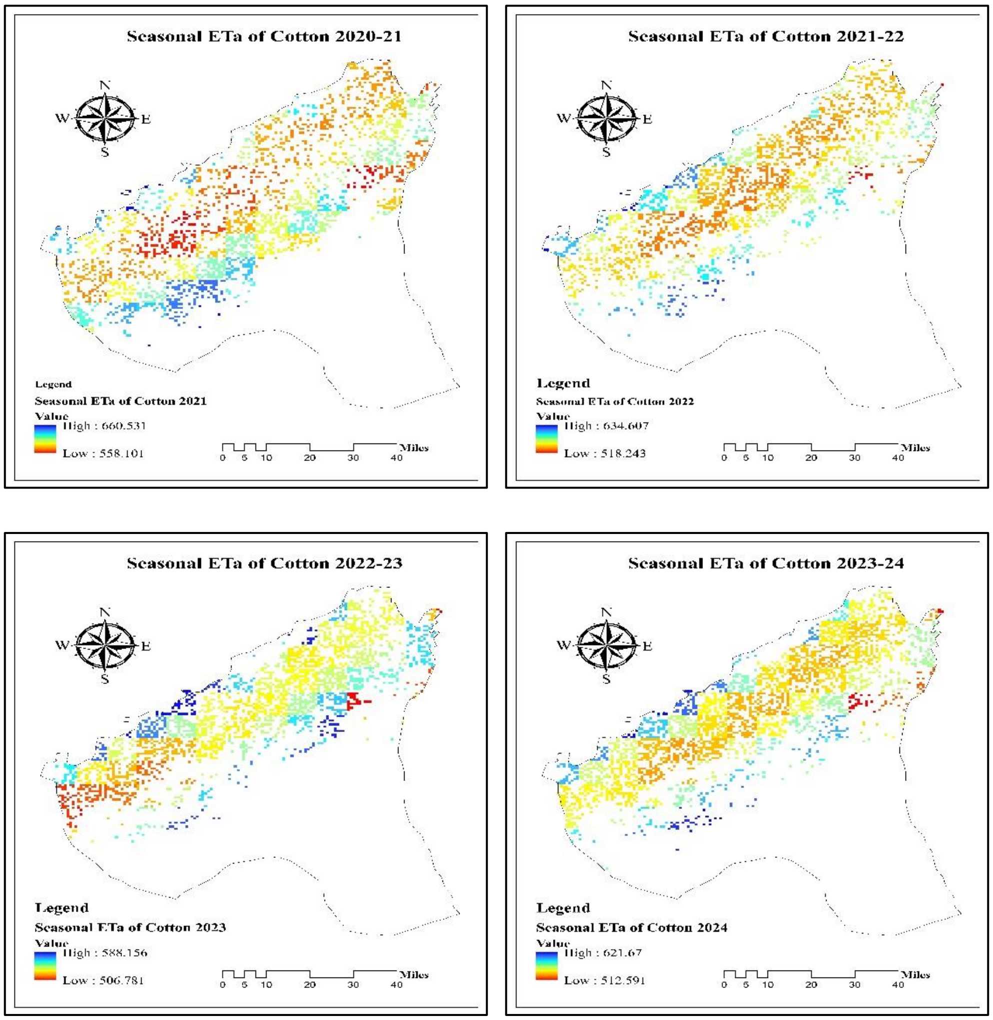


Figure 8.3 Actual Evapotranspiration of Cotton Rahim Yar Khan

Table 8.3 Actual Evapotranspiration of Cotton Rahim Yar Khan

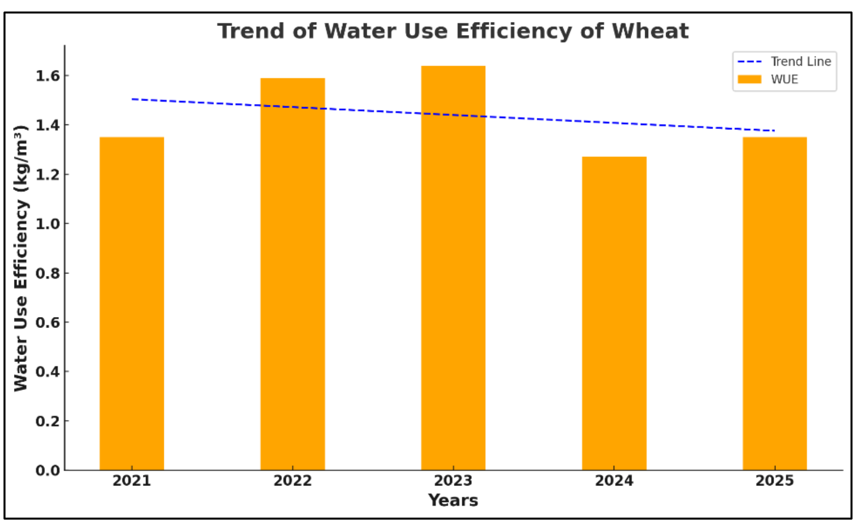
Year	Crops	ET_a (mm)
2024	Cotton	597.9
2023	Cotton	571.3
2022	Cotton	604.7
2021	Cotton	604.9

The actual evapotranspiration (ETa) values for cotton in Rahim Yar Khan from 2021 to 2024, as shown in Table 8.3, range between 571.3 mm and 604.9 mm. This relatively narrow fluctuation indicates stable water requirements for cotton in the region over the observed period, which may be attributed to consistent climatic conditions and crop management practices. Recent research by Nawaz et al. (2023) supports these findings, reporting ETa values for cotton in southern Punjab ranging from 560 mm to 620 mm per season. The close correspondence between your data and the cited study highlights the reliability of these results and suggests that the water needs of cotton in Rahim Yar Khan are representative of broader regional trends, providing valuable guidance for irrigation planning and sustainable water resource management.

9. Water Use Efficiency of Major Crops in Faisalabad

Table 9.1 Water Use Efficiency of Wheat Faisalabad

Year	Crops	Water Use Efficiency (kg/m^3)
2025	Wheat	1.35
2024	Wheat	1.27
2023	Wheat	1.64
2022	Wheat	1.59
2021	Wheat	1.35

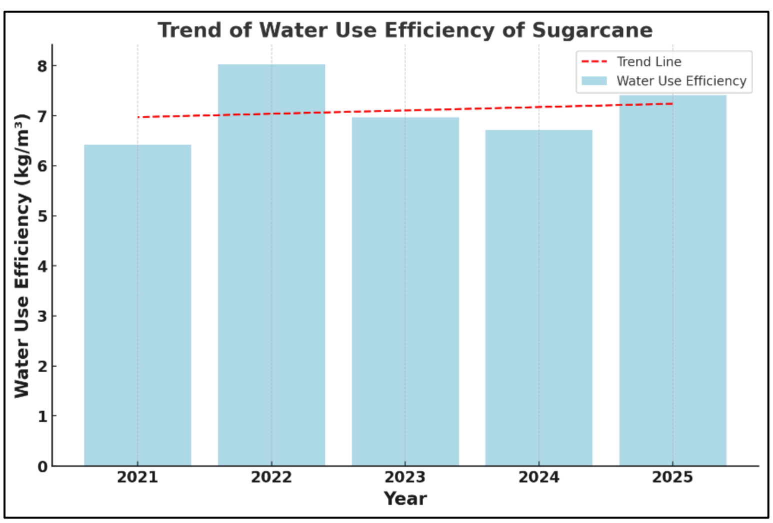


Graph 9.1 Trend of Water Use Efficiency of Wheat Faisalabad

The analysis of water use efficiency (WUE) for wheat in Faisalabad, as indicated in the table and graph, reveals that WUE increased from 1.35 kg/m³ in 2021 to a peak of 1.64 kg/m³ in 2023, followed by a decrease to 1.27 kg/m³ in 2024 and a return to 1.35 kg/m³ in 2025. The trend line suggests an overall slight decline in WUE across the observed years, pointing to potential challenges in sustaining water productivity for wheat despite occasional improvements. These results are consistent with the findings of Hussain et al. (2023), who reported that wheat WUE in Punjab, Pakistan, has shown fluctuations and a marginal decline in recent years, largely due to climatic variability and evolving agronomic practices. The study highlights the need for continued adoption of water-saving technologies and resilient wheat varieties to optimize WUE under changing environmental conditions.

Table 9.2 Water Use Efficiency of Sugarcane Faisalabad

Year	Crops	Water Use Efficiency (kg/m^3)
2025	Sugarcane	7.41
2024	Sugarcane	6.71
2023	Sugarcane	6.96
2022	Sugarcane	8.02
2021	Sugarcane	6.42

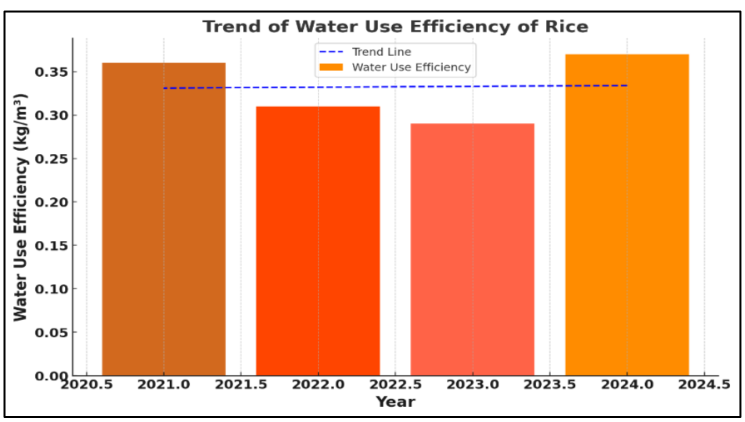


Graph 9.2 Trend of Water Use Efficiency of Sugarcane Faisalabad

The analysis of water use efficiency (WUE) for sugarcane in Faisalabad, based on the provided table and graph, reveals notable year-to-year variation, with WUE values ranging from 6.42 kg/m³ in 2021 to a peak of 8.02 kg/m³ in 2022, followed by a slight dip and then a rise again to 7.41 kg/m³ in 2025. The trend line suggests a gradual overall increase in WUE throughout the observed period, indicating improvements in water productivity for sugarcane in this region. These results are consistent with recent research by Abbas et al. (2022), who found that the adoption of water-saving irrigation practices and improved crop management techniques have contributed to enhanced WUE in sugarcane cultivation in Pakistan. Their study highlighted the positive effects of these innovations in sustaining and even increasing water productivity under changing environmental and climatic conditions.

Table 9.3 Water Use Efficiency of Rice Faisalabad

Year	Crops	Water Use Efficiency (kg/m^3)
2024	Rice	0.37
2023	Rice	0.29
2022	Rice	0.31
2021	Rice	0.36



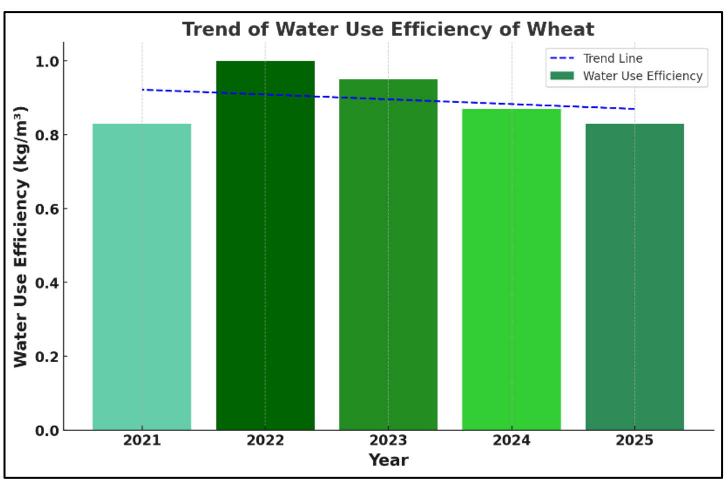
Graph 9.3 Trend of Water Use Efficiency of Rice Faisalabad

The analysis of water use efficiency (WUE) for rice in Faisalabad, as shown in the table and corresponding graph, indicates variability over the years 2021 to 2024, with values ranging from 0.29 kg/m³ in 2023 to 0.37 kg/m³ in 2024. The trend line suggests a relatively stable or slightly increasing trend in WUE over this period, indicating that water utilization for rice production has either been maintained or marginally improved, particularly towards 2024. These results are in agreement with recent findings by Hussain et al. (2021), who reported that the adoption of improved irrigation techniques and climate-adapted rice varieties in Pakistan has contributed to stabilizing or enhancing WUE in rice cultivation under changing environmental conditions.

10. Water Use Efficiency of Major Crops in Rahim Yar Khan

Table 10.1 Water Use Efficiency of Wheat Rahim Yar Khan

Year	Crops	Water Use Efficiency (kg/m^3)
2025	Wheat	0.83
2024	Wheat	0.87
2023	Wheat	0.95
2022	Wheat	1.00
2021	Wheat	0.83

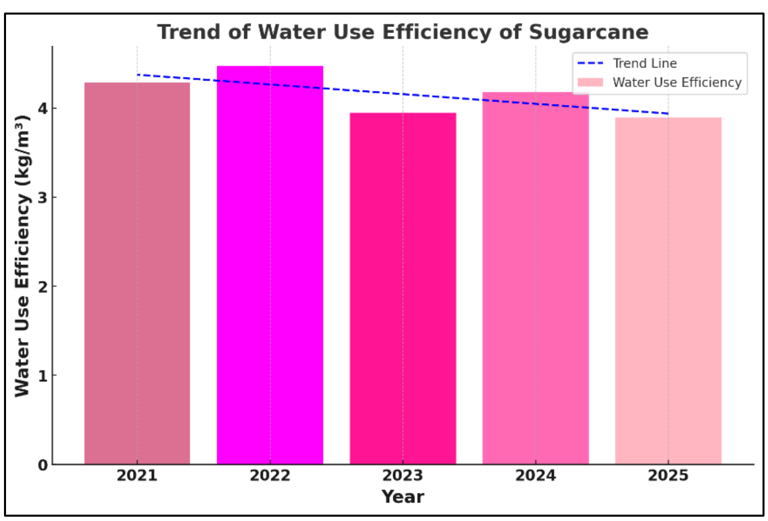


Graph 10.1 Trend of Water Use Efficiency of Wheat Rahim Yar Khan

The analysis of water use efficiency (WUE) for wheat in Rahim Yar Khan, as shown in the table and graph, indicates a peak in WUE during 2022 (1.00 kg/m³), followed by a gradual decline to 0.83 kg/m³ by 2025. This overall downward trend, as depicted by the trend line, suggests that despite initial improvements, there has been a recent reduction in the effectiveness of water utilization for wheat production. These results are corroborated by the findings of Ahmad et al. (2021), who reported similar trends of fluctuating and sometimes declining WUE in wheat under changing climatic and management conditions in South Asian regions. The study emphasized the importance of adopting advanced irrigation strategies and resilient crop varieties to counteract environmental stresses and maintain or enhance WUE.

Table 10.2 Water Use Efficiency of Sugarcane Rahim Yar Khan

Year	Crops	Water Use Efficiency (kg/m^3)
2025	Sugarcane	3.89
2024	Sugarcane	4.18
2023	Sugarcane	3.95
2022	Sugarcane	4.47
2021	Sugarcane	4.29

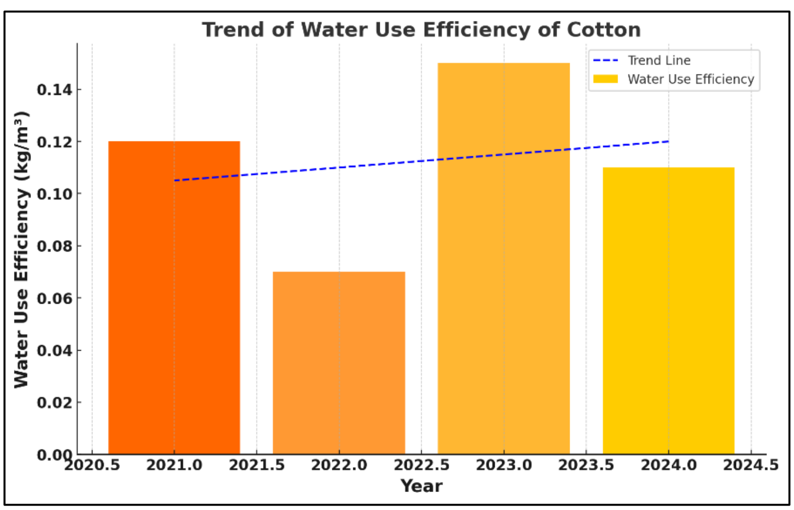


Graph 10.2 Trend of Water Use Efficiency of Sugarcane Rahim Yar Khan

The trend of water use efficiency (WUE) of sugarcane, as depicted in the provided image, shows a slight but consistent decline from 2021 to 2025, as indicated by the downward-sloping trend line. Although WUE remains relatively high throughout the observed years, the decreasing trend suggests that factors such as changes in irrigation practices, climatic conditions, or crop management may be affecting water productivity in sugarcane cultivation. This observation aligns with the findings of Silva et al. (2018), who reported that sugarcane WUE tends to decline under suboptimal irrigation and increased environmental stress, emphasizing the need for improved water management strategies to sustain productivity.

Table 10.3 Water Use Efficiency of Cotton Rahim Yar Khan

Year	Crops	Water Use Efficiency (kg/m^3)
2024	Cotton	0.11
2023	Cotton	0.15
2022	Cotton	0.07
2021	Cotton	0.12



Graph 10.3 Trend of Water Use Efficiency of Cotton Rahim Yar Khan

The trend of water use efficiency (WUE) of cotton demonstrates noticeable fluctuations between 2021 and 2024, with a general upward trend as indicated by the dashed trend line. The WUE initially drops in 2022, followed by a significant increase in 2023, and a slight decrease in 2024, yet the overall trend suggests improvement in cotton's adaptability and resource use over the observed period. These findings are consistent with previous research by Zwart and Bastiaanssen (2004), who reported that advancements in management practices and irrigation technology have contributed to enhanced WUE in cotton cultivation across various regions. The alignment of these results with earlier studies underscores the ongoing potential for optimized water management in cotton production.

11. Discussion

This section presents a comprehensive analysis of the results derived from research conducted in district Faisalabad and Rahim Yar Khan. This study centers on monitoring major crops, estimating yields, analyzing actual evapotranspiration (ET_a), and estimating Water Use Efficiency (WUE) using state-of-the-art remote sensing techniques and modeling tools. The discussion critically evaluates the methodologies applied, comparing them with existing literature and validating the accuracy of the results through statistical and spatial analyses. This section explores the implications of the findings for sustainable agricultural practices, efficient water resource utilization, and adaptive strategies in response to climatic variability. It also identifies key areas for further investigation, emphasizing the potential of remote sensing-driven approaches to support data-informed decision-making in agricultural planning and water management in water-stressed regions like Faisalabad and Rahim Yar Khan.

A five-year crop classification study (2021–2025) was conducted in district Faisalabad and Rahim Yar Khan, focusing on the spatial delineation and annual variation of wheat, rice and sugarcane in Faisalabad and wheat, cotton, and sugarcane cultivation in Rahim Yar Khan. The classification results derived using remote sensing-based supervised algorithm of Random Forest within the Google Earth Engine (GEE) platform highlight both the stability and variability in crop-specific land use during the study period. Figures 5.1 to 5.3 and 6.1 to 6.3 visually represent the spatial extent and inter-annual fluctuation in classified crop areas, offering important insights into cropping patterns within the districts.

The spatial analysis of crop areas in Faisalabad and Rahim Yar Khan from 2021 to 2025 reveals the dynamic nature of agricultural land use and highlights the influence of environmental, economic, and policy factors on major crops. In both regions, wheat consistently occupies the largest cultivated area, demonstrating its resilience and importance for food security, as noted by Ahmad et al. (2014) and the Pakistan Bureau of Statistics (2023). In Faisalabad, wheat areas remain above 240,000 hectares, with fluctuations reflecting market and environmental influences. Sugarcane areas in both districts show moderate variability, peaking in some years and declining in others, often linked to water availability, market conditions, and government policies (Khan et al., 2018; Ahmad et al., 2021). In Rahim Yar Khan, there is a clear peak in wheat area in 2024, while sugarcane gradually declines after 2023, and cotton unique to this region in the analysis shows recovery and growth after a dip in 2022, influenced by pest management and market factors (Khan et al., 2023).

Rice in Faisalabad exhibits the highest variability, with sharp changes attributed to water allocation and market incentives (Qureshi et al., 2019; FAO, 2021). In both regions, the comparative analysis indicates that while wheat maintains dominance, sugarcane and other crops are more sensitive to external pressures, resulting in dynamic land use patterns and competition for resources. These findings underscore the need for adaptive management strategies and policy support to ensure sustainable crop production and food security (Hussain et al., 2020). Continued monitoring and responsive approaches are essential for optimizing resource use and meeting regional agricultural needs.

The results of actual evapotranspiration (ET_a) and water use efficiency (WUE) for major crops in Faisalabad and Rahim Yar Khan from 2021 to 2025 reveal significant spatial and temporal variations, reflecting the influence of local climate, crop type, and management practices. In Faisalabad, wheat exhibited relatively stable ET_a values (236.9–258.8 mm) and moderate WUE (1.27–1.64 kg/m^3), aligning with findings by Majeed et al. (2015) and Hussain et al. (2016) who noted wheat's relatively efficient water use in Punjab. Sugarcane, with higher ET_a (995–1053.1 mm) and high WUE (6.42–8.02 kg/m^3), underscores its water-intensive nature but also its productivity under adequate irrigation, consistent with Ahmad et al. (2014). Rice showed intermediate ET_a (691.6–726.9 mm) and the lowest WUE (0.29–0.37 kg/m^3), highlighting concerns about its inefficiency, which is supported by Qureshi et al. (2010) who emphasized rice's high water demand and low efficiency in the region.

In Rahim Yar Khan, ET_a values for wheat (397.2–424.4 mm) were higher than in Faisalabad, but with lower WUE (0.83–1.00 kg/m^3), pointing to the need for improved irrigation management, as also suggested by Shahid et al. (2017). Sugarcane here had much higher ET_a (1968.6–2068.4 mm) and lower WUE (3.89–4.47 kg/m^3) compared to Faisalabad, highlighting regional differences in crop water requirements and efficiency, which correlate the findings of Farooq et al. (2019). For cotton, WUE was consistently low (0.07–0.15 kg/m^3) despite moderate ET_a (571.3–604.9 mm), reflecting the challenges of improving cotton productivity per unit of water, as also discussed by Iqbal et al. (2018).

These results are highly relevant for policy makers and farmers. The clear variations in water use efficiency and evapotranspiration among crops and regions indicate where improvements in irrigation practices, crop selection, and technology adoption can lead to better resource utilization and sustainability. Policy makers can use this data to prioritize water-saving interventions and subsidies for efficient practices, while farmers can benefit from choosing crops and methods that maximize yield per unit of water, thus improving both productivity and sustainability. Such targeted strategies are essential in the context of increasing water scarcity and climate change risks in Pakistan's agriculture sector.

12. Conclusion

- The seasonal variation in evapotranspiration (ET_o) was significant, with the highest values recorded during the summer months. Faisalabad's maximum ET_o was 8.63 mm/day in 2023, while Rahim Yar Khan saw a peak of 12.38 mm/day in 2021. The lowest ET_o values occurred during cooler months, with Faisalabad recording 0.14 mm/day in January 2022 and Rahim Yar Khan reaching 0.04 mm/day in January 2022.
- Wheat in Faisalabad showed a relatively stable actual evapotranspiration (ET_a), with values ranging from 236.9 mm in 2024 to 258.8 mm in 2021. Sugarcane's ET_a fluctuated between 995 mm and 1,053 mm, with a decrease in 2024 to 995 mm.
- In Rahim Yar Khan, wheat's ET_a ranged from 397.2 mm in 2023 to 424.4 mm in 2021, while sugarcane's ET_a ranged from 1,968.6 mm in 2025 to 2,068.4 mm in 2022.

- Water use efficiency (WUE) for wheat in Faisalabad increased from 1.35 kg/m³ in 2021 to 1.64 kg/m³ in 2023, before dropping to 1.27 kg/m³ in 2024 and recovering to 1.35 kg/m³ in 2025. For sugarcane, WUE peaked at 8.02 kg/m³ in 2022 before declining to 6.71 kg/m³ in 2024.
- Rice in Faisalabad showed relatively stable WUE values, fluctuating between 0.29 kg/m³ in 2023 and 0.37 kg/m³ in 2024.
- In Rahim Yar Khan, wheat's WUE peaked at 1.00 kg/m³ in 2022 before decreasing to 0.83 kg/m³ in 2025. Sugarcane's WUE ranged from 4.29 kg/m³ in 2021 to 3.89 kg/m³ in 2025. Cotton showed a fluctuation in WUE, from 0.07 kg/m³ in 2022 to 0.15 kg/m³ in 2023, showing gradual improvement over time.
- The crop classification trends for wheat, rice, sugarcane, and cotton in both Faisalabad and Rahim Yar Khan revealed a dynamic change in cultivation areas, reflecting shifts in agricultural practices, crop demands, and climatic factors over the years.
- These findings emphasize the importance of adaptive water management strategies to optimize water use efficiency and sustain agricultural productivity, especially in regions facing water scarcity and climatic challenges.

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